

ASIA TECHNOLOGY

ABF price hikes to outpace our estimates, with solid GM expansion from 2Q26; revise up TPs for all our TW ABF substrate players by 81% to 112%

Following up on our ABF upgrade report from early this year (see [here](#)), we are seeing ABF substrate demand grow much faster than our original expectation. **We now expect the ABF TAM to expand at a 62% 2025-28E CAGR to US\$33bn in 2028, up from US\$7.7bn in 2025 (Exhibit 3, vs. our prior estimate of a 33% CAGR). Moreover, we expect the S/D gap for ABF substrates to widen from an 8% undersupply in 2026 to 34/51% in 2027/28 (Exhibit 1 vs. 5%/21%/42% undersupply in 2026/27/28 previously),** driven by stronger than expected demand from AI servers and general server CPU substrates, which together accounted for 26% of ABF industry demand in 2025, and should increase to 41/53/62% of demand in 2026/27/28; [Exhibit 4](#)). As a result, we expect the ABF substrate LTA/Spot prices to increase by 10-15%/20%+ QoQ per quarter from 3Q26 through at least the end of 2027, with shortages likely extending into 2028 or possibly beyond as ABF lead times have lengthened from 3-4 months at the start of the year to more than 12 months currently.

Given our bullish view on the ABF substrate pricing/profitability outlook, **we raise our TPs for NYPCB/Kinsus/Unimicron/ZDT to NT\$2,310/1,120/1,140/824 from NT\$1,115/555/630/388 with 4-22%/10-38% upward earnings revisions in 2027/28 and introduce our 2029 earnings estimates.** Maintain our Buy rating on NYPCB (on CL)/Kinsus/ZDT, but stay Neutral on Unimicron.

ABF substrates could become a key bottleneck for the overall AI/server industry, supporting a solid price hike outlook

We began to see NYPCB/Kinsus UTR pick up rapidly ([Exhibit 2](#)) in 1H26, following Ibiden's nearly full utilization for GPU applications from 2H24 and Unimicron's full utilization from 2H25, and started seeing spot price hikes from April, which implies that shortages of substrate supply likely began much earlier than our original expectation, as we had expected them to emerge only from 2H26, thanks to the stronger than anticipated demand.

Our latest industry checks suggest that the ABF substrate lead times have now reached 12 months (vs. only 3-4 months as of early 2026), with more customers asking for ABF substrate capacity beyond 2029. Moreover, we started to see some tier 2/3 ABF substrate suppliers selling their on-hand capacity based on bidding, suggesting a much more favorable pricing/profitability outlook for the spot price for

Chao Wang
+886(2)2730-4195 | kuan-
chao.wang@gs.com
Goldman Sachs (Asia) L.L.C., Taipei
Branch

Daiki Takayama
+81(3)4587-9870 |
daiki.takayama@gs.com
Goldman Sachs Japan Co., Ltd.

Allen Chang
+852-2978-2930 |
allen.k.chang@gs.com
Goldman Sachs (Asia) L.L.C.

Al Wang
+886(2)2730-4081 | al.wang@gs.com
Goldman Sachs (Asia) L.L.C., Taipei
Branch

Goldman Sachs does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. For Reg AC certification and other important disclosures, see the Disclosure Appendix, or go to www.gs.com/research/hedge.html. Analysts employed by non-US affiliates are not registered/qualified as research analysts with FINRA in the U.S.

ABF substrate suppliers.

All the above industry signs and data-point suggests to us that the ABF substrate shortage could last for more than 30 months, and **our supply demand analysis suggests the shortage ratio of ABF substrates could reach 14%/34%/51% in 2H26/2027/2028, indicating potential for an even faster pace of price increases for ABF substrates.** We expect the market to eventually recognize that substrate supply constraints could impact the overall AI server ramp up speed. **Overall, we expect substrate LTA/spot prices to increase 10-15%/20%+ QoQ from 3Q26 through to the end of 2027, suggesting a solid upside potential to our GM estimate for all 4 ABF substrate players in Taiwan.**

AI ASIC servers and general server CPUs are the key drivers of stronger than expected demand growth

We now see the overall ABF substrate TAM expanding at a 62% 2025-28E CAGR vs. our original expectation of a 33% 2025-28E CAGR, with the stronger than expected demand growth mainly driven by **(1) the stronger than expected AI ASIC substrate volume/pricing growth, and (2) much more favorable than expected server CPU substrate demand growth.** We expect AI server demand (25% of ABF demand in 2026E) to grow at a 101% 2025-28E CAGR, benefiting Ibiden/Unimicron the most, as the two companies together still account for 74% of the AI server share in 2026. We also expect general server demand (15% of ABF demand in 2026E) to grow at a 47% 2025-28E CAGR, benefiting Kinsus the most, as server CPUs could account for 40%+ of the company's total ABF revenue in 2028, thanks to rising demand for high computing performance ICs and IC substrate technology migration.

For the other applications, we expect switches to grow the fastest, at a 20% 2025-28E CAGR, driven by solid expansion in ABF substrate size and layer count following the adoption of 1.6T/3.2T switch ICs starting in 2H26/2028. This should benefit NYPCB the most given the company's current 40%+ ABF revenue exposure to high-end networking ICs.

2Q26 preview and 3Q26 outlook: Spot price player to strongly outperform with a significant jump in GM/OPM

For the upcoming 2Q results, we believe all four TW ABF substrate companies will report better than expected earnings, 4-38% above BBG consensus, mainly driven by much stronger ABF substrate shipment volumes and better ASP/pricing. We also expect **key spot price supplier, NYPCB, to enjoy the strongest GM/OPM expansion, from 15.8%/12.1% in 1Q26 to 26.3%/23.2% in 2Q26 (vs. consensus of 19.8%/17.5%),** supported by the 40%+ spot price hikes in 2Q26.

Going into 3Q, we expect all four TW ABF substrate suppliers to deliver even stronger earnings growth, while **our 3Q26 operating income estimates for all 4 ABF substrate suppliers are 9-53% above BBG consensus.** We expect **the 4 TW ABF players' GM/OPM to expand by 3.5ppt/4.7ppt in 3Q26,** considering the even more positive ABF substrate ASP/price hike trend from 3Q26. **For NYPCB, we believe GM/OPM will again expand the most in 3Q26, by +6.7ppt/+7.2ppt GM/OPM QoQ,** further supported by our more favorable ABF substrate spot price outlook.

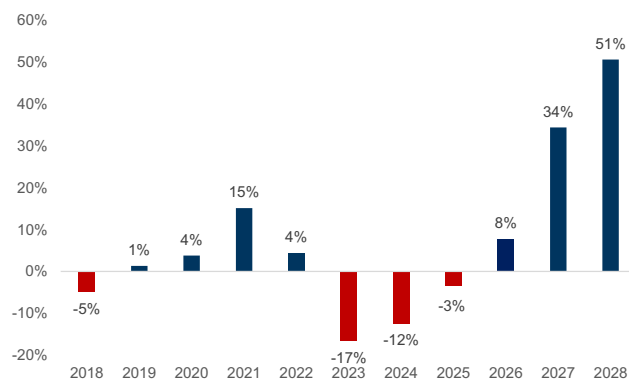
ABF substrate supply/demand gap to widen to 14%/34%/51% in 2H26/2027/2028, with a much stronger pricing outlook

We revise up our ABF substrate TAM 2025-28E CAGR from 33% to 62%, driven by (1) our faster than expected AI IC substrate demand outlook, mainly from ASIC volume growth, as well as solid progress on specification upgrades, and (2) solid demand for agentic AI server CPUs. We also see NVDA's launch of the new Vera standalone rack suggesting further upside demand potential for server CPU substrate suppliers in the coming years.

On the other hand, we are starting to see all ABF substrate players' capacity located in Japan/Korea/Taiwan become fully utilized, thanks to the solid demand outlook and the comparatively slower capacity expansion plans from key suppliers. Also, we have started to see ABF substrate lead times reach 12+ months currently vs. 3-4 months at the beginning of this year, which further echos our view on the solid demand growth since 2Q25.

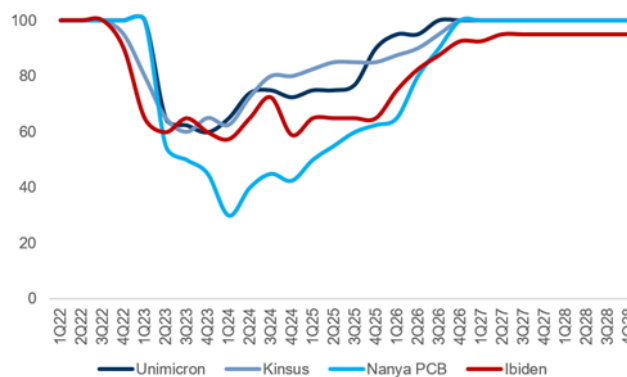
As such, we now expect the overall ABF substrate supply/demand gap to reach 8/34/51% in 2026/27/28 (vs. our original expectation at 5%/21%/42%), while we see the faster than expected demand ramp up from April 2026, creating a supply shortage 1 quarter ahead of our estimate—we originally expected a shortage to begin in 2H26, with spot price product pricing hikes, but the shortage actually started in early 2Q26. The ABF substrate supply shortage beginning from April this year has directly led to a 40-60% pricing increase in the spot price market in 2Q26. We believe the pricing outlook for spot ABF substrates may accelerate in 2H26, as the shortage ratio likely rises further from 2% in 1H26 to 14% in 2H26E. Heading into 2027/28, we believe ABF substrate supply could be the IC supply chain bottleneck, potentially suggesting an even stronger pricing outlook for ABF substrates, as most all AI related customers have been focusing more on time to market timing than cost control. Accordingly, we believe the spot price of ABF substrates is poised to rise further by 60-80% YoY in both 2027 and 2028.

Exhibit 1: We expect the ABF industry S/D gap to continue expanding in coming years
Supply shortage ratio



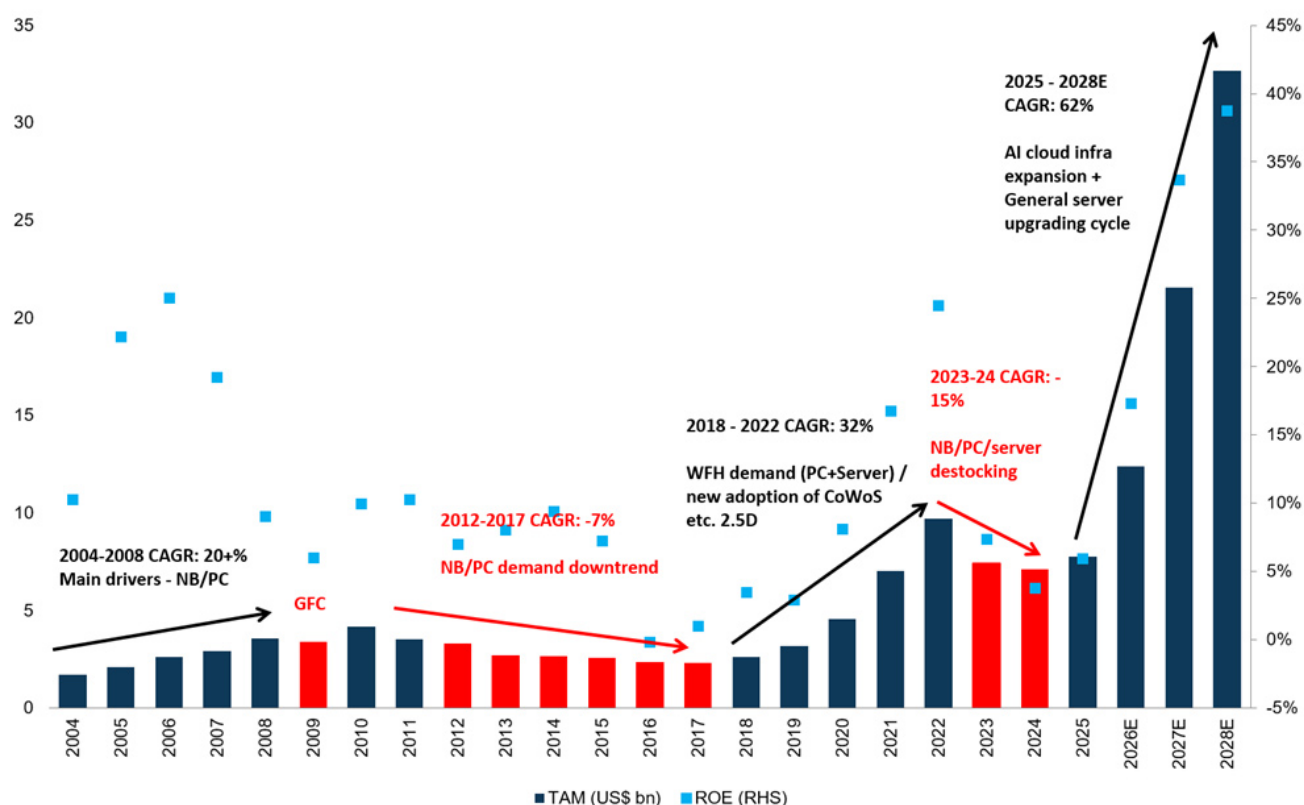
Source: Company data, Goldman Sachs Global Investment Research

Exhibit 2: We believe the ABF substrate industry UTR will continue trending up in 2H26 and reach full UTR



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 3: We believe we are in the early stages of a new ABF industry upcycle, and expect to see new ROE highs in coming years

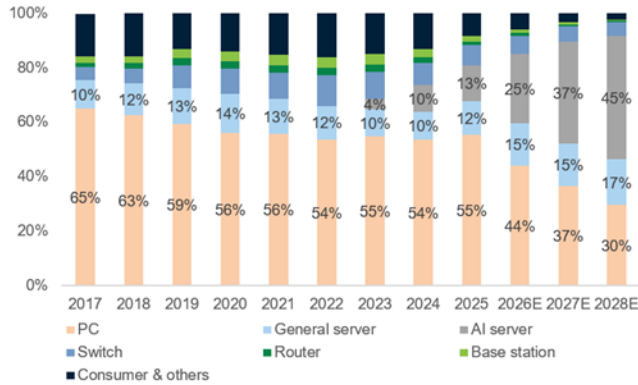


Source: Company data, Goldman Sachs Global Investment Research

Demand update: AI servers and server CPUs are the key drivers the ABF substrate demand

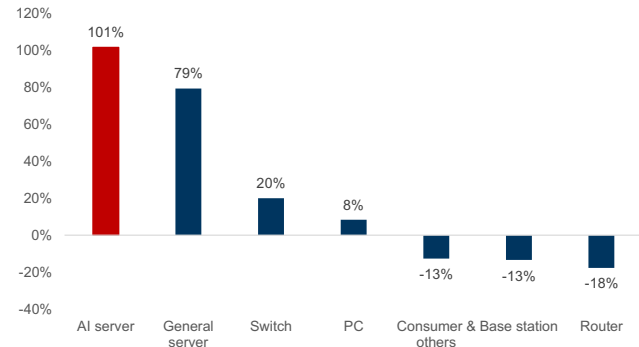
We expect ABF substrate TAM to reach US\$33bn by 2028, up from US\$7.7bn in 2025, implying a 62% 2025-28E CAGR (vs. our prior 33% estimate). Our upward revision is mainly due to (1) strong AI ABF substrate demand, driven primarily by a stronger ASIC substrate volume/price growth than we had expected, and (2) solid demand from general server CPUs, driven by agentic AI. We expect AI servers/general servers to account for 45%/17% of total 2028E demand, up from 13%/12% in 2025 ([Exhibit 4](#)), and expect AI servers/general server ABF demand to grow at a 101%/47% 2025-28E CAGR ([Exhibit 5](#)).

Exhibit 4: ABF demand by application - AI server contribution continues to increase



Source: Company data, Goldman Sachs Global Investment Research

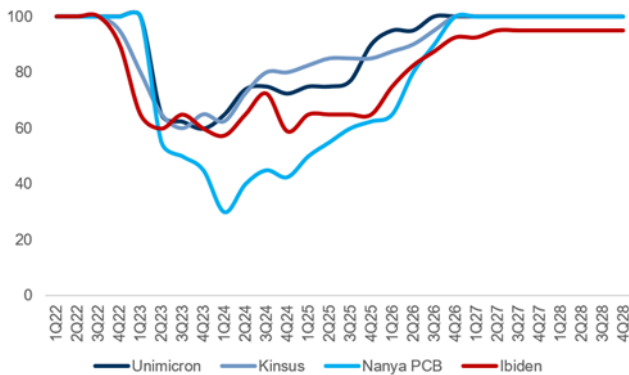
Exhibit 5: AI servers to see strong 2025-28E demand growth, with a CAGR of 101%, followed by general servers at 79%



Source: Company data, Goldman Sachs Global Investment Research

The substrate supply shortage started much earlier than our original expectation (we had expected that shortages would only start from 2H26), which is reflected in (1) increasing UTR, as NYPCB and Kinsus have already seen their UTRs trend up in 1H26, and we expect all major ABF suppliers to operate at full UTR from 2H26 through at least 2028; and (2) lengthening lead times, which have now extended to 12+ months, vs. 3-4 months in early 2026, and are likely to move closer to the 18-30 months range observed during the previous upcycle in 2020-22.

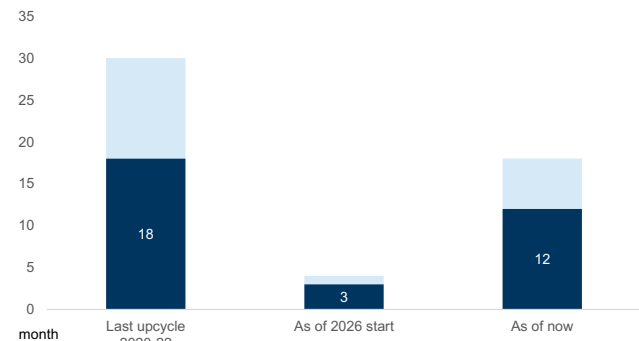
Exhibit 6: We believe the ABF substrate industry UTR will continue trending up in 2H26 and reach full UTR



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 7: ABF substrate lead times have now reached 12+ months vs. 3-4 months at the start of 2026

Lead time lower end and upper end



Source: Company data, Goldman Sachs Global Investment Research

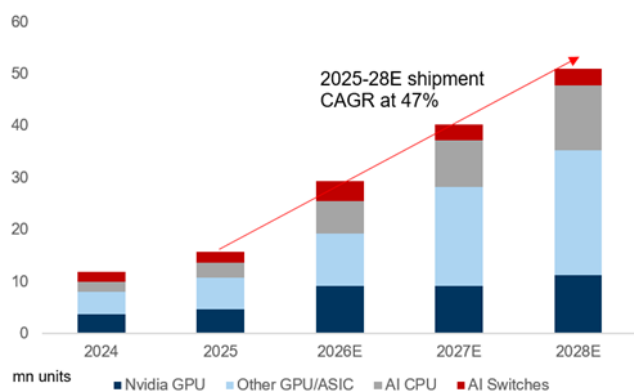
We continue to see higher layer counts and larger body sizes for ABF substrates used in general server CPUs and AI chips, resulting in higher ABF area consumption per package. The ABF specification upgrade for general server CPUs is relatively moderate, and we expect a 30%+ increase in ABF area consumption for the next generation of Intel/AMD CPUs. However, we expect AI GPU's ABF area consumption to increase the most significantly. For example, we expect Nvidia's Rubin GPU to consume 123% more ABF area compared with the current Blackwell GPU, while the corresponding AI CPU is expected to see a 47% increase in ABF area consumption when comparing Nvidia's Vera and Grace CPUs. Current ASIC and switch platform upgrades also imply more ABF area consumption, but we expect their increase to fall behind that of AI GPUs.

Exhibit 8: Higher ABF area consumption driven by rising layer count and larger body size

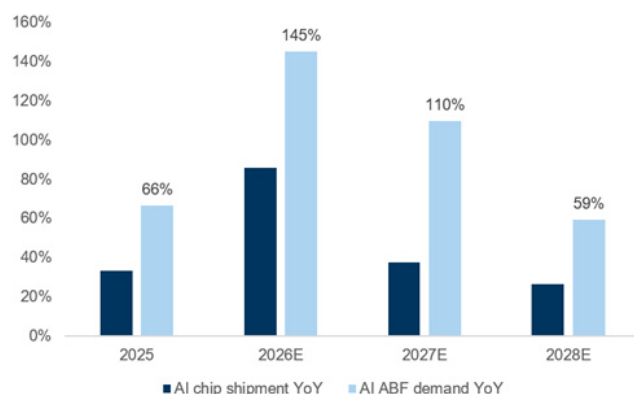
Company	Type	Chip	Layer count	Length (mm)	Width (mm)	Body size (mm ²)	ABF area consumption
Intel	Server CPU	Eagle Stream	16	100.0	56.5	5,650	90,400
		Birch Stream	18	104.5	70.5	7,367	132,611
		Oak Stream	20	109.2	88.0	9,606	192,129
AMD	Server CPU	Genoa	18	72.0	75.4	5,429	97,718
		Turin	18	72.0	75.4	5,429	97,718
		Venice	20	80.1	83.8	6,711	134,224
Nvidia	AI CPU	Grace	14	68.0	73.0	4,964	69,496
		Vera	18	76.0	75.0	5,700	102,600
Nvidia	AI GPU	Hopper	14	60.0	55.0	3,300	46,200
		Blackwell	16	80.0	70.0	5,600	89,600
		Rubin	20	100.0	100.0	10,000	200,000
Amazon	ASIC	Trainium 2	14	87.5	72.5	6,344	88,813
		Trainium 3	16	100.0	87.5	8,750	140,000
		Trainium 4	18	120.0	100.0	12,000	216,000
Broadcom	Switch	Tomahawk 4	12	77.5	77.5	6,006	72,075
		Tomahawk 5	16	87.5	87.5	7,656	122,500
		Tomahawk 6	18	110.0	120.0	13,200	237,600

Source: Company data, Goldman Sachs Global Investment Research

We expect AI ABF demand to grow at a 101% 2025-28E CAGR and account for 45% of total ABF demand by 2028, up from 13% in 2025, supported by strong shipment growth across all AI chips, especially for ASICs. For Nvidia GPUs, ABF demand growth is increasing driven by higher ABF content per chip, outpacing that of other AI chips. While we also expect increasing ABF content per chip for ASICs, ABF demand growth from ASICs over 2025-28E will likely be driven more by shipment volume growth, at a 58% 2025-28E CAGR. For AI CPUs, we also see NVDA's launch of the new Vera standalone rack implying more demand upside for AI CPU shipments. Overall, the increasing ABF area consumption for next gen AI chips should continue to drive overall ABF demand, and we expect AI ABF demand YoY growth to outpace AI chip shipment YoY growth in 2025-28E (Exhibit 10).

Exhibit 9: AI chip shipments to grow at a 47% CAGR from 2025-28E

Source: Company data, Goldman Sachs Global Investment Research

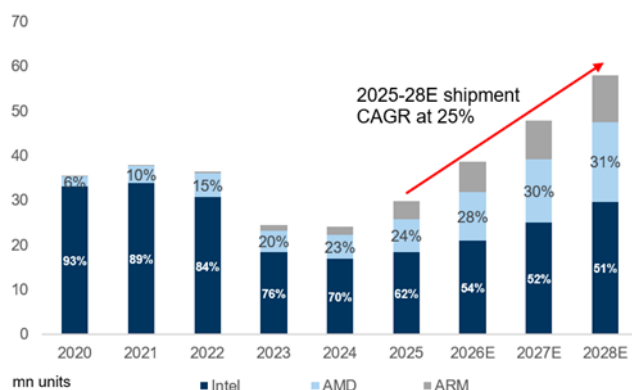
Exhibit 10: AI ABF demand growth YoY should significantly outpace shipments growth YoY

Source: Company data, Goldman Sachs Global Investment Research

We expect general server CPU ABF demand to grow at a 47% 2025-28E CAGR and account for 17% of total ABF demand by 2028, up from 12% in 2025, with stable CPU shipments growing at a 25% 2025-28E CAGR. General server CPU demand is primarily driven by the growing adoption of agentic AI, which we could see lasting for another 3-5 years. As agentic AI applications become mainstream, we expect the GPU/CPU attach rates to move from 2x to 1.1-1.4x over time, implying that each GPU deployment will

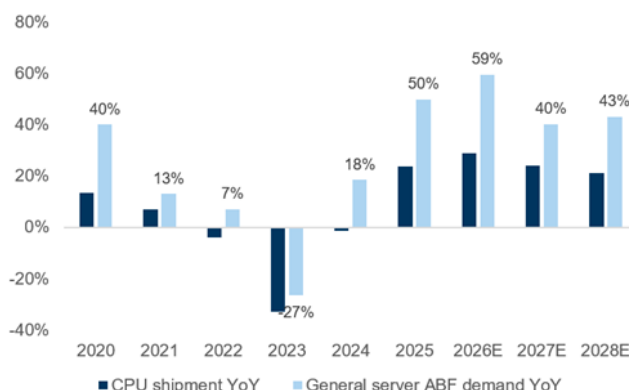
require more CPUs (see GS analyst Jim Schneider's [Intel initiation](#)). We expect a modest increase in ABF area consumption for general server CPUs compared with AI chips, with ABF area consumption increasing by 30%+ for next generation of Intel/AMD CPUs. As a result, we continue to expect general server CPU ABF demand YoY growth to outpace general server CPU shipments YoY in 2025-28E ([Exhibit 12](#)).

Exhibit 11: General server CPU shipments to grow at a moderate 25% CAGR from 2025-28E



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 12: General server ABF demand YoY will continue to outpace shipments YoY



Source: Company data, Goldman Sachs Global Investment Research

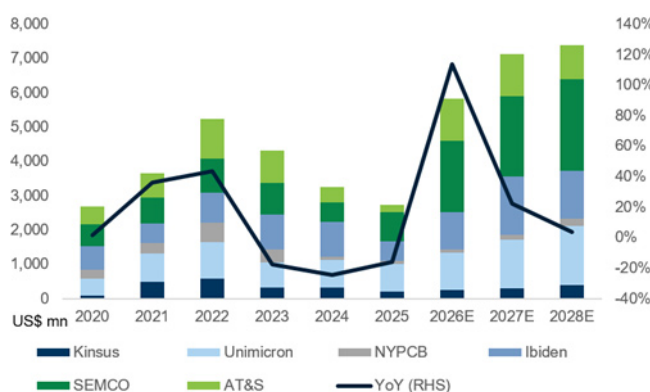
Supply update: We expect ABF suppliers will start to announce more CAPEX plans in coming quarters, although new capacity will likely only come online after 2H28

Compared with our previous ABF report update, we have seen some Taiwan ABF suppliers revise up their 2026 CAPEX plans since then: ZDT raised its plan from NT\$50bn to NT\$80bn in April and Unimicron raised theirs from NT\$25bn to NT\$34bn on their 1Q26 results call). Considering ongoing upgrades in ABF substrate specifications (including higher layer count, larger body size, etc.) and ABF suppliers continuing to invest in technology upgrades to support next generation packaging (including CoPoS, glass substrate, EMIB-T, etc.), customers are requesting more long-term capacity, with some already requesting capacity beyond 2029. As a result, we expect ABF suppliers to announce additional CAPEX plans over the coming quarters/years to support sustained demand growth.

However, ABF substrate capacity volume will continue to decline as packaging body sizes increase and layer counts rise. This capacity inefficiency is driven by increasing production complexity associated with more advanced substrate designs; hence, we expect existing capacity efficiency to continue declining by ~10% annually if specification upgrades continue at their current pace.

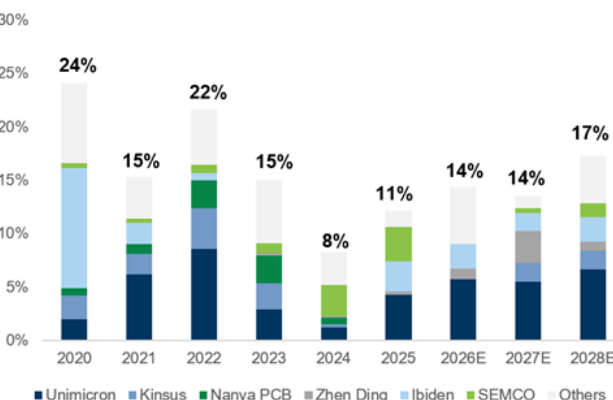
We expect overall ABF supply to reach a 15% CAGR in 2025-28E ([Exhibit 14](#)), supported by elevated CAPEX plans across the major ABF suppliers growing at 40% CAGR in 2025-28E ([Exhibit 13](#)). The current pace of capacity expansion is tracking behind CAPEX increases mainly due to (1) long equipment lead times, with some newly announced capacity becoming available only after 2028 and (2) some suppliers are also allocating CAPEX to other segment of their businesses (including MLCC, PCB, etc). However, we are seeing more ABF suppliers planning capacity beyond 2029, as their customers request more long-term capacity ([Exhibit 15](#)).

Exhibit 13: We expect ABF suppliers' combined CAPEX to remain elevated in order to meet growing capacity requirements



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 14: We believe the overall pace of ABF substrate industry capacity will increase gradually in the coming years



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 15: ABF suppliers' capacity expansion plan going forward

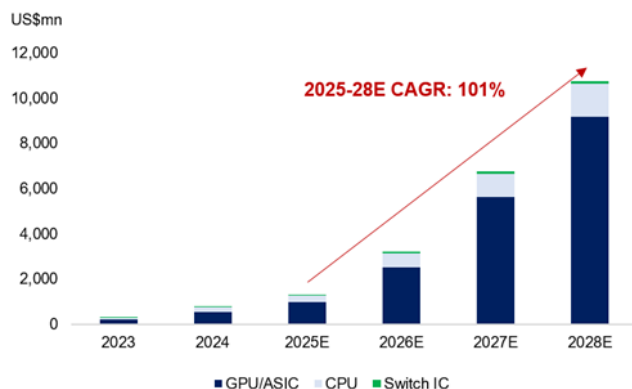
Company	Expansion plant	Capex	Time of new capacity come on-line	Capacity expansion
Unimicon	KF plant 2 YM plant 2	KF plant 1: NT\$30bn+ KF plant 2: NT\$35-40bn YM plant 2: NT\$35-40bn	KF plant 1 phase 3: starts in 3Q26 KF plant 2: start in 2H27 YM plant 2: start in mid 2028	KF plant 1 phase 3: Additional 15-20% of total capacity per GSe KF plant 2: Additional 15-20% of total capacity per GSe YM plant 2: Additional 10-15% of total capacity per GSe
Kinsus	Yangmei K6 plant	Yangmei K6 plant phase 2: NT\$7bn in 2026 Yangmei K6 plant phase 3: NT\$8bn in 2027 per GSe Yangmei K6 plant phase 4: NT\$8bn+ in 2028 per GSe	Yangmei K6 plant phase 2: 2027 Yangmei K6 plant phase 3: 2028 Yangmei K6 plant phase 4: 2029	Yangmei K6 plant phase 2: 10mn units Yangmei K6 plant phase 3: 15mn units Yangmei K6 plant phase 4: 15mn units
Ibiden	Gama Plant Ono Plant	Gama Plant: ~220bn JPY Ono Plant: ~280bn JPY	Gama Plant: From FY27, Mass Production From FY28 Ono Plant: 1/4 each from FY25-FY28 (280bn investment is for latter half)	Gama Plant: ~15% vs FY24 ASIC/AI Server Capacity Ono Plant: ~85% vs FY24 ASIC/AI Server Capacity
AT&S	Kulim plant	Kulim Phase 2: EUR1.5-2.0bn (supported by LT customer commitments)	Kulim phase 2: 2028 per GSe	Kulim Phase 2: additional 25% of total capacity per GSe
ZDT	Kaohsiung plant Shenzhen plant	Capex for IC substrate at NT\$12-16bn in 2026	Shenzhen fab 2: equipment installation in 2H26/2027, MP in 2027 Kaohsiung AI fab: MP in 2H26	Kaohsiung: Additional 38% capacity expansion per GSe
TOPPAN	AST Phase1 (Singapore) Ishikawa (Advanced packaging) New Ishikawa FC-BGA AST Phase2 (Singapore)	N/A	AST Phase1 line: Mass production launch scheduled for FY27 Ishikawa (Advanced packaging): Statup schedules for July 2026, mass production scheduled for FY28 or later New Ishikawa FC-BGA line: Mass production scheduled for FY30 or later AST Phase2 line: Timing TBD	Ishikawa (Advanced packaging): limited contribution to the overall capacity Others: roughly the same size as the existing three domestic lines
SEMCO	Vietnam plant	KRW1.8bn	Vietnam plant: operation starts in 2H27	Vietnam plant: additional 15% of total capacity per GSe

Source: Company data, Goldman Sachs Global Investment Research

Supply / demand update: We believe the overall ABF substrate industry will see shortage ratios of 14%/34%/51% in 2H26/2027/2028, leading to a favorable pricing outlook

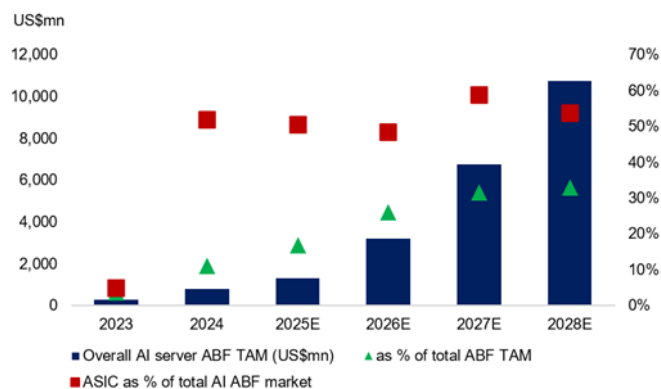
We update our ABF industry model, and now expect the ABF TAM to grow at a 62% 2025-28E CAGR (vs. 33% over 2025-28E, previously). The upward revision is mainly driven by (1) the strong AI server growth, especially on the ASIC shipments, and (2) rapid growth in general server CPU shipments which we expect to continue ramping at a 25% 2025-28E CAGR, driven mainly by solid agentic AI demand. Furthermore, while several suppliers have announced new capacity expansion plan for the coming years and some ABF substrate suppliers are even evaluating new CAPEX plans for the long term (beyond 2029), we expect near term supply to remain constrained by long equipment lead times, resulting in overall ABF supply growing at only a 15% 2025-28E CAGR, leading to overall undersupply ratios of 14%/34%/51% in 2H26/27/28, respectively.

Exhibit 16: We forecast key AI IC substrate TAM will continue to grow at a 101% 2025-28E CAGR (revised up from 66%)



Source: Company data, Goldman Sachs Global Investment Research

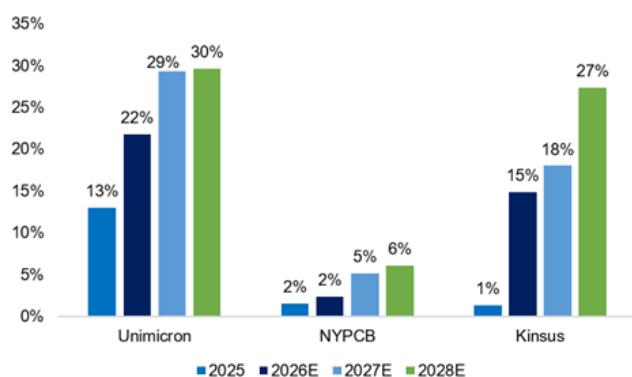
Exhibit 17: ASIC to account for 50%+ of total AI ABF substrate demand in 2028E



Source: Company data, Goldman Sachs Global Investment Research

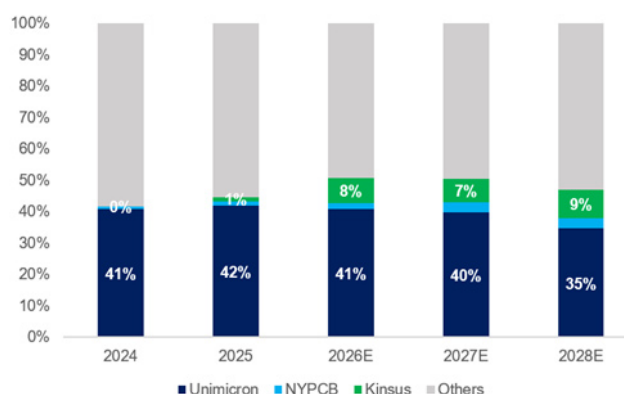
In terms of market share, we continue to see Kinsus gaining market share in the NVDA supply chain (especially for AI server CPUs and NVSwitch ICs). We expect Kinsus to hold 8/7/9% of AI substrate market share in 2026/27/28E (up from 1% in 2025E) and AI to account for 15%/18%/27% of the company's total revenue in 2026/27/28E (vs. 2% in 2025).

Exhibit 18: We believe Kinsus' AI ABF revenue exposure will accelerate in 2025-28E



Source: Company data, Goldman Sachs Global Investment Research

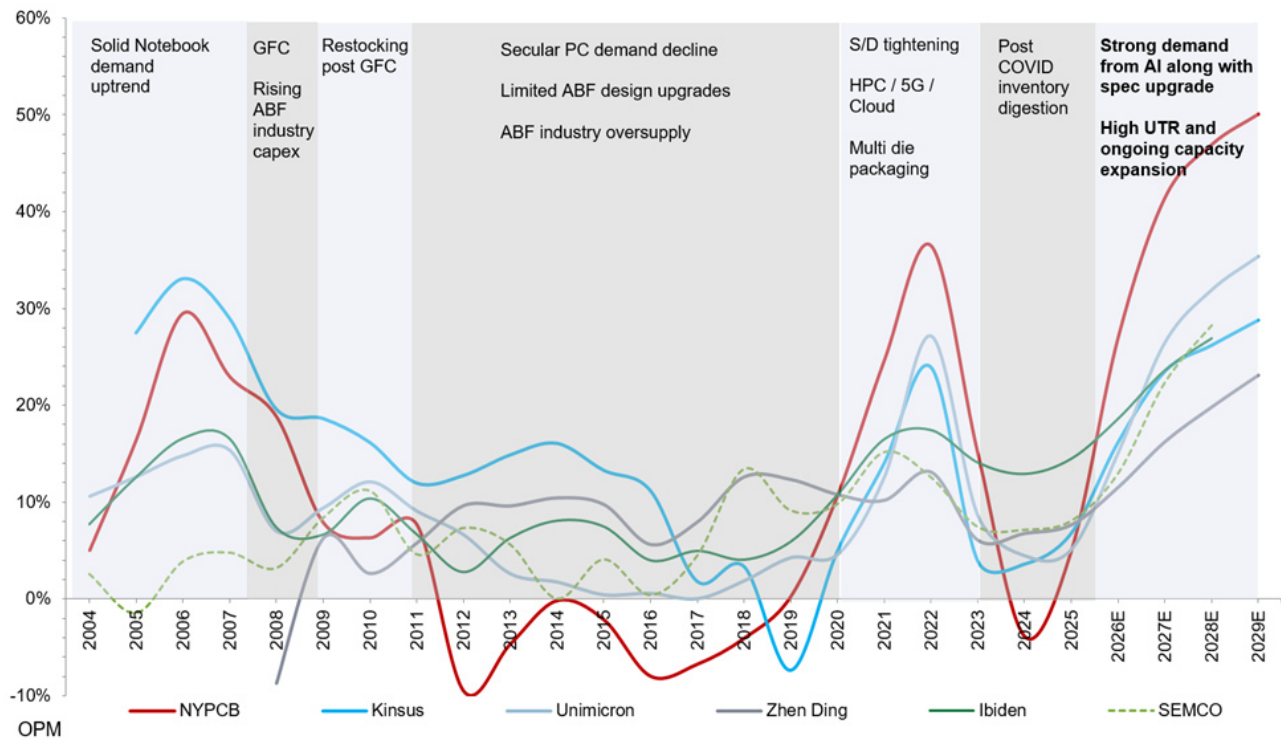
Exhibit 19: We expect Kinsus will continue to gain share in the AI ABF market in coming years



Source: Company data, Goldman Sachs Global Investment Research

Due to the accelerating shortage ratio we anticipate in 2025-28E, we now expect the ABF substrate LTA/spot prices to increase by 10-15%/20%+ QoQ per quarter from 3Q26 through to at least the end of 2027, and expect the shortages will continue into 2028 and possibly beyond. The stronger ABF pricing environment suggests a solid upside potential to our OPM estimates for all 4 ABF substrate players in Taiwan in the coming quarters/years ([Exhibit 20](#)).

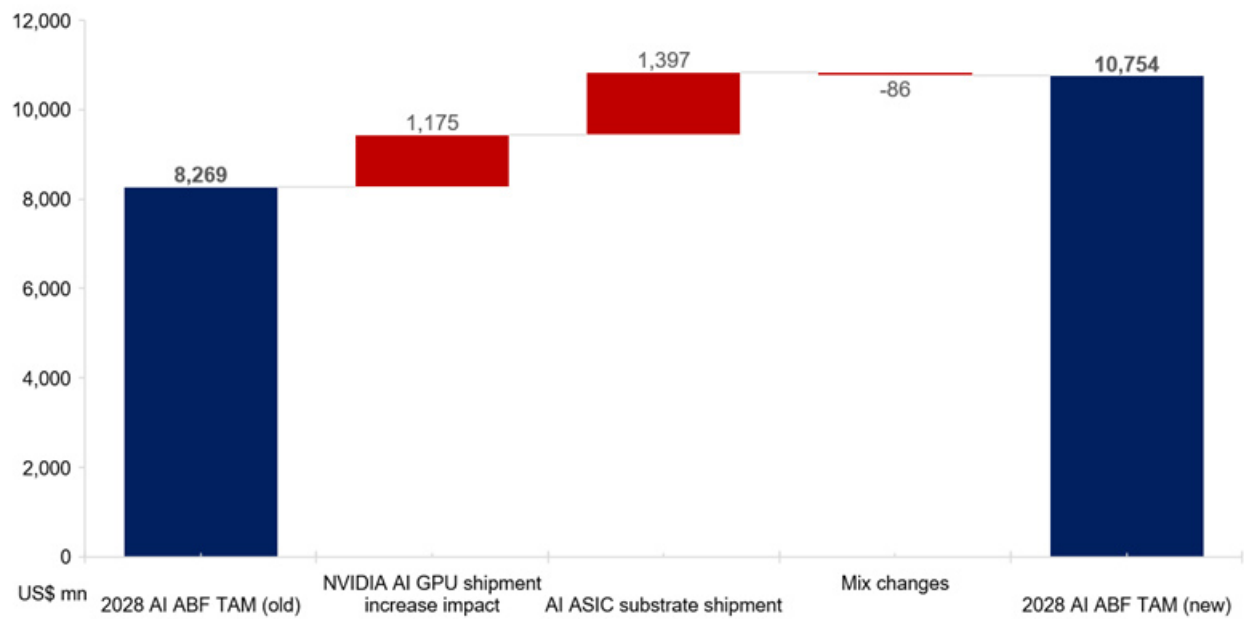
Exhibit 20: We expect the ongoing OPM uptrend to continue, while we believe spot price players could deliver their strongest OPM growth in coming quarters/years



OPMs are on a companywide basis

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 21: We revise up our 2028E AI ABF substrate TAM by 30% to factor in our more positive view on stronger AI substrate shipments and larger AI substrate sizes



Source: Company data, Goldman Sachs Global Investment Research

2Q26 & 3Q26 preview - stronger ABF substrate pricing to drive GM expansion

For the upcoming 2Q result, **we believe all four TW ABF substrate companies will see better than expected earnings (4-38% higher than BBG consensus)**, mainly driven by much higher ABF substrate volume shipments and stronger ASP/pricing thanks to the substrate supply tightness from 2Q.

Considering that spot prices increased by 40%+ from April, we expect key spot price player NYPCB, which had 80% exposure to the spot price market in 1Q26, to see its GM expand from <16% in 1Q26 to 26%+ in 2Q26. The 10ppt QoQ GM expansion should mainly be due to better pricing, likely allowing NYPCB to continue to outgrow all other ABF substrate suppliers who have greater LTA exposure. For Unimicron/Kinsus/ZDT, we expect the three companies' GM to surpass BBG consensus by 0.2/1.2/0.3ppt, respectively, on strengthening product mix and improving pricing conditions.

Going into 3Q, we expect all four TW ABF substrate suppliers to deliver even stronger earnings growth, while **our 3Q26 operating income estimate for all 4 ABF substrate suppliers are 9-53% higher than BBG consensus**. We expect the **4 TW ABF players' GM/OPM to expand by 3.5ppt/4.7ppt in 3Q26**, reflecting even more positive ABF substrate ASP/price hike trends in 3Q26. **For NYPCB, we believe the GM/OPM will again expand the most in 3Q26 with +6.7ppt/+7.2ppt GM/OPM QoQ**, again supported by a more favorable ABF substrate spot price outlook. For Unimicron/Kinsus/ZDT, we expect these three companies' OPM to surpass BBG consensus by 1.5/3.7/0.9ppt, on a stronger product mix and improved pricing conditions.

Overall, **we believe the ABF substrate pricing uptrend just started in 2Q26, and we should see further acceleration in ABF substrate pricing given the shortage conditions (i.e., no IC can be packaged without a substrate), leading to a favorable profitability outlook over the coming quarters/years**. Within the Taiwan ABF substrate industry, we prefer the spot price players (with NYPCB (CL-Buy) as the key spot price player here), considering the potential upside to their GM/OPM outlook, which are both still 15ppt below the peak level. We believe sustained AI infrastructure investment demand and long equipment lead times will continue to drive spot prices much more strongly than LTA pricing, suggesting solid earnings upside potential, which we should start seeing in 2Q26 results.

Exhibit 22: We believe NYPCB will start seeing a major jump in GM from 2H26, while all our covered ABF players should see better GMs in coming quarters

4 TW ABF substrate players' quarterly GM trend



Source: Company data, Goldman Sachs Global Investment Research

Exhibit 23: 4 TW ABF substrate players' 3Q26 vs 2Q26 earnings estimate, we expect solid margin expansion for NYPCB/Kinsus

NT\$ mn; GSe	NYPCB			Kinsus			Unimicron			ZDT		
	2Q26	3Q26	QoQ	2Q26	3Q26	QoQ	2Q26	3Q26	QoQ	2Q26	3Q26	QoQ
Sales	13,575	17,659	30%	12,496	14,756	18%	42,920	50,138	17%	48,431	63,831	32%
Gross Profit	3,568	5,815	63%	3,282	4,533	38%	9,306	12,435	34%	10,430	15,637	50%
EBIT	3,144	5,354	70%	1,717	2,814	64%	5,743	8,474	48%	4,013	8,488	112%
Net Income	2,755	4,316	57%	1,093	2,065	89%	5,339	6,225	17%	2,418	5,393	123%
EPS (NT\$)	4.26	6.68	57%	2.34	4.42	89%	3.47	4.05	17%	2.26	5.03	123%
GM (%)	26.3%	32.9%	6.7ppt	26.3%	30.7%	4.5ppt	21.7%	24.8%	3.1ppt	21.5%	24.5%	3.0ppt
OPM (%)	23.2%	30.3%	7.2ppt	13.7%	19.1%	5.3ppt	13.4%	16.9%	3.5ppt	8.3%	13.3%	5.0ppt

Source: Company data, Goldman Sachs Global Investment Research

Exhibit 24: 2Q26 & 3Q26 GSe vs. BBG consensus comp table

NT\$ mn	2Q26E				3Q26E			
	NYPCB	Kinsus	Unimicron	ZDT	NYPCB	Kinsus	Unimicron	ZDT
Sales	13,575	12,496	42,920	48,431	17,659	14,756	50,138	63,831
Gross Profit	3,568	3,282	9,306	10,430	5,815	4,533	12,435	15,637
EBIT	3,144	1,717	5,743	4,013	5,354	2,814	8,474	8,488
Net Income	2,755	1,093	5,339	2,418	4,316	2,065	6,225	5,393
EPS (NT\$)	4.26	2.34	3.47	2.26	6.68	4.42	4.05	5.03
GM (%)	26.3%	26.3%	21.7%	21.5%	32.9%	30.7%	24.8%	24.5%
OPM (%)	23.2%	13.7%	13.4%	8.3%	30.3%	19.1%	16.9%	13.3%
BBG consensus	NYPCB	Kinsus	Unimicron	ZDT	NYPCB	Kinsus	Unimicron	ZDT
Sales	13,318	12,529	42,338	45,704	16,449	14,451	48,869	62,737
Gross Profit	2,633	3,136	9,082	9,711	3,834	3,892	11,446	15,367
EBIT	2,329	1,602	5,285	3,821	3,498	2,225	7,542	7,802
Net Income	1,953	1,077	5,034	2,323	2,879	1,600	6,348	4,899
EPS (NT\$)	3.10	2.24	3.28	2.11	4.58	3.16	4.11	4.50
GM (%)	19.8%	25.0%	21.5%	21.2%	23.3%	26.9%	23.4%	24.5%
OPM (%)	17.5%	12.8%	12.5%	8.4%	21.3%	15.4%	15.4%	12.4%
Differences (GSe vs. BBG)	NYPCB	Kinsus	Unimicron	ZDT	NYPCB	Kinsus	Unimicron	ZDT
Sales	2%	0%	1%	6%	7%	2%	3%	2%
Gross Profit	35%	5%	2%	7%	52%	16%	9%	2%
EBIT	35%	7%	9%	5%	53%	26%	12%	9%
Net Income	41%	1%	6%	4%	50%	29%	-2%	10%
EPS (NT\$)	38%	4%	6%	7%	46%	40%	-2%	12%
GM (%)	6.5ppt	1.2ppt	0.2ppt	0.3ppt	9.6ppt	3.8ppt	1.4ppt	0.0ppt
OPM (%)	5.7ppt	1.0ppt	0.9ppt	-0.1ppt	9.1ppt	3.7ppt	1.5ppt	0.9ppt

Source: Company data, Goldman Sachs Global Investment Research, Bloomberg

Earnings estimate changes: Faster than expected margin expansion driven by better ABF substrate

pricing

Earnings revisions

NYPCB

We revise up our 2026/27/28E earnings estimates by 9%/22%/36% to factor in much more favorable supply/demand outlook for ABF substrate, with higher pricing driven by supply constraints. The stronger-than-expected demand pull-in for ABF substrates should continue to support pricing uplift longer term. Considering the stronger ABF substrate pricing outlook, we revise up our 2026/27/28E GM by 4.8/8.3/10.5ppt. We revise down 2026E revenue by 12% due to the weaker-than-expected shipments driven by material shortages the in near term but revise up our 2028E revenue by 9% to factor in stronger long-term demand for ABF substrates.

Also, we introduce our 2029E earnings estimates in this report.

Exhibit 25: NYPCB earnings revision table

NYPCB P&L (NT\$ mn)	2026E New	2026E Old	Diff.	2027E New	2027E Old	Diff.	2028E New	2028E Old	Diff.	2029E New
Sales	64,805	73,477	-12%	127,361	127,932	0%	194,486	178,551	9%	295,639
Gross Profit	19,351	18,378	5%	55,812	45,427	23%	95,782	69,255	38%	155,062
EBIT	17,488	16,279	7%	52,764	42,363	25%	91,270	65,114	40%	148,104
Net Income	14,478	13,232	9%	41,783	34,166	22%	71,191	52,209	36%	116,387
EPS (NT\$)	22.41	20.48	9%	64.66	52.88	22%	110.17	80.80	36%	180.12
Ratio analysis										
Gross margin	29.9%	25.0%	4.8pp	43.8%	35.5%	8.3pp	49.2%	38.8%	10.5pp	52.4%
EBIT margin	27.0%	22.2%	4.8pp	41.4%	33.1%	8.3pp	46.9%	36.5%	10.5pp	50.1%
Net margin	22.3%	18.0%	4.3pp	32.8%	26.7%	6.1pp	36.6%	29.2%	7.4pp	39.4%

Source: Goldman Sachs Global Investment Research

Kinsus

We revise down our 2026E net earnings by 2% after applying a more conservative margin assumption on its IC substrate business in 2Q26E but revise up our 2027/28E net earnings by 8/38%, driven by a better ABF substrate pricing outlook and a much more favorable supply demand outlook for the overall ABF substrate market. We revise up our 2027/28 revenue estimate by 9%/34% and revise up our 2028 GM by 0.5ppt.

Also, we introduce our 2029E earnings estimates in this report.

Exhibit 26: Kinsus earnings revision table

Kinsus P&L (NT\$ mn)	2026E New	2026E Old	Diff.	2027E New	2027E Old	Diff.	2028E New	2028E Old	Diff.	2029E New
Sales	54,949	55,190	0%	95,645	87,849	9%	154,185	114,962	34%	236,684
Gross Profit	15,529	15,580	0%	33,138	30,449	9%	57,734	42,497	36%	94,697
EBIT	8,895	8,864	0%	22,494	20,669	9%	40,422	29,592	37%	68,211
Net Income	6,250	6,358	-2%	17,030	15,702	8%	31,701	22,986	38%	54,488
EPS (NT\$)	13.36	13.60	-2%	36.42	33.58	8%	67.79	49.15	38%	116.52
Ratio analysis										
Gross margin	28.3%	28.2%	0.0ppt	34.6%	34.7%	0.0ppt	37.4%	37.0%	0.5ppt	40.0%
EBIT margin	16.2%	16.1%	0.1ppt	23.5%	23.5%	0.0ppt	26.2%	25.7%	0.5ppt	28.8%
Net margin	11.4%	11.5%	-0.1ppt	17.8%	17.9%	-0.1ppt	20.6%	20.0%	0.6ppt	23.0%

Source: Goldman Sachs Global Investment Research

Unimicron

We revise down our 2026E revenue by 1% to reflect weaker-than-expected shipments due to material shortages but revise up our 2027/28E revenue estimate by 2/6% to factor in the better demand visibility for ABF substrates, supporting a steeper pricing

uplift in the longer term. We revise up our 2026/27/28E GM by 0.8/2.4/2.1ppt, resulting in 7/12/10% higher EPS estimates with improvement in operating leverage.

Also, we introduce our 2029E earnings estimates in this report.

Exhibit 27: Unimicron earnings revision table

Unimicron P&L (NT\$ mn)	2026E New	2026E Old	Diff.	2027E New	2027E Old	Diff.	2028E New	2028E Old	Diff.	2029E New
Sales	188,516	191,063	-1%	299,174	293,972	2%	424,900	402,458	6%	583,347
Gross Profit	44,333	43,497	2%	101,274	92,603	9%	166,786	149,463	12%	246,555
EBIT	28,373	27,336	4%	79,287	70,995	12%	135,710	120,028	13%	206,613
Net Income	25,490	23,874	7%	61,401	54,844	12%	102,742	93,527	10%	162,368
EPS (NT\$)	16.58	15.53	7%	40.33	36.03	12%	66.82	60.83	10%	106.66
Ratio analysis										
Gross margin	23.5%	22.8%	0.8pp	33.9%	31.5%	2.4pp	39.3%	37.1%	2.1pp	42.3%
EBIT margin	15.1%	14.3%	0.7pp	26.5%	24.2%	2.4pp	31.9%	29.8%	2.1pp	35.4%
Net margin	13.5%	12.5%	1.0pp	20.5%	18.7%	1.9pp	24.2%	23.2%	0.9pp	27.8%

Source: Goldman Sachs Global Investment Research

ZDT

We revise down our 2026E revenue by 1% considering the material shortage, which has led to weaker-than-expected shipments and revise up our 2027/28E revenue estimates by 2/11% to factor in the stronger-than-expected ABF substrate pull-ins and a better pricing outlook. We revised down ZDT's 2026E GM by 0.2ppt due to weaker shipments, resulting in 3% lower 2026E EPS estimates, but we revise up 2027/28 GM by 0.4/2.3ppt to reflect a much stronger than expected ABF substrate pricing outlook, resulting in 4/25% higher EPS estimates.

Also, we introduce our 2029E earnings estimates in this report

Exhibit 28: ZDT earnings revision table

ZD Tech P&L (NT\$ mn)	2026E New	2026E Old	Diff.	2027E New	2027E Old	Diff.	2028E New	2028E Old	Diff.	2029E New
Sales	224,680	226,449	-1%	327,764	321,782	2%	454,047	407,834	11%	602,605
Gross Profit	53,140	54,060	-2%	85,019	82,163	3%	133,376	110,548	21%	197,093
EBIT	25,882	26,593	-3%	53,206	50,922	4%	89,985	71,561	26%	139,326
Net Income	16,363	16,789	-3%	33,322	31,958	4%	54,498	43,500	25%	83,856
EPS (NT\$)	15.27	15.66	-3%	31.09	29.81	4%	50.84	40.58	25%	78.23
Ratio analysis										
Gross margin	23.7%	23.9%	-0.2pp	25.9%	25.5%	0.4pp	29.4%	27.1%	2.3pp	32.7%
EBIT margin	11.5%	11.7%	-0.2pp	16.2%	15.8%	0.4pp	19.8%	17.5%	2.3pp	23.1%
Net margin	7.3%	7.4%	-0.1pp	10.2%	9.9%	0.2pp	12.0%	10.7%	1.3pp	13.9%

Source: Goldman Sachs Global Investment Research

Valuation

NYP CB

We are Buy rated (on CL) with a new 12-month TP of NT\$2,310 (from NT\$1,115), based on the same 21.0x P/E on 2028E EPS (rolled over from 2027E). We believe the company could trade at 1.5x STDV above the past upcycle average P/E of 13.3x, as we continue to expect better pricing condition for ABF substrates, and factor in our latest earnings revisions.

Exhibit 29: NYPCB P&L table

NT\$m	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026E	2027E	2028E	2029E
Revenue	8,458	9,583	10,967	11,165	11,177	13,575	17,659	22,393	25,209	29,637	34,098	38,417	32,283	40,173	64,805	127,361	194,486	295,639
Gross profit	430	769	1,050	1,398	1,771	3,568	5,815	8,197	10,106	12,457	15,202	18,047	358	3,647	19,351	55,812	95,782	155,062
Operating expense	(397)	(403)	(429)	(435)	(419)	(424)	(461)	(559)	(635)	(706)	(808)	(899)	(1,624)	(1,665)	(1,863)	(3,048)	(4,511)	(6,958)
Operating income	33	366	620	963	1,352	3,144	5,354	7,638	9,471	11,751	14,394	17,147	(1,267)	1,982	17,488	52,764	91,270	148,104
Pretax income	256	(227)	907	1,409	1,598	3,444	5,534	7,818	9,471	11,751	14,394	17,147	163	2,346	18,394	52,764	91,270	148,824
Taxes expense	(49)	39	(182)	(207)	(290)	(689)	(1,217)	(1,720)	(1,800)	(2,585)	(3,167)	(3,429)	41	(399)	(3,916)	(10,981)	(20,079)	(32,437)
Net income	207	(187)	725	1,202	1,309	2,755	4,316	6,098	7,672	9,166	11,228	13,718	204	1,947	14,478	41,783	71,191	116,387
EPS, NT\$	0.32	(0.29)	1.12	1.86	2.03	4.26	6.68	9.44	11.87	14.19	17.38	21.23	0.32	3.01	22.41	64.66	110.17	180.12

Ratio analysis and assumption																		
As % of sales																		
Gross margin	5.1%	8.0%	9.6%	12.5%	15.8%	26.3%	32.9%	36.6%	40.1%	42.0%	44.6%	47.0%	1.1%	9.1%	29.9%	43.8%	49.2%	52.4%
Operating expense ratio	4.7%	4.2%	3.9%	3.9%	3.8%	3.1%	2.6%	2.5%	2.5%	2.4%	2.4%	2.3%	5.0%	4.1%	2.9%	2.4%	2.3%	2.4%
Operating margin	0.4%	3.8%	5.7%	8.6%	12.1%	23.2%	30.3%	34.1%	37.6%	39.6%	42.2%	44.6%	-3.9%	4.9%	27.0%	41.4%	46.9%	50.1%
Net margin	2.5%	-2.0%	6.6%	10.8%	11.7%	20.3%	24.4%	27.2%	30.4%	30.9%	32.9%	35.7%	0.6%	4.8%	22.3%	32.8%	36.6%	39.4%
QoQ growth (%)																		
Revenue	7.5%	13.3%	14.4%	1.8%	0.1%	21.5%	30.1%	26.8%	12.6%	17.6%	15.1%	12.7%	-	-	-	-	-	-
Gross profit	2441.0%	78.7%	36.5%	33.2%	26.7%	101.4%	63.0%	41.0%	23.3%	23.3%	22.0%	18.7%	-	-	-	-	-	-
Operating income	-108.5%	1006.2%	69.4%	55.2%	40.4%	132.5%	70.3%	42.7%	24.0%	24.1%	22.5%	19.1%	-	-	-	-	-	-
Net income	16.1%	-190.3%	-486.7%	65.8%	8.9%	110.5%	56.7%	41.3%	25.8%	19.5%	22.5%	22.2%	-	-	-	-	-	-
YoY growth (%)																		
Revenue	19.1%	18.0%	19.3%	41.9%	32.1%	41.7%	61.0%	100.6%	125.5%	118.3%	93.1%	71.6%	-24%	24%	61%	97%	53%	52%
Gross profit	-213.1%	319.0%	95.2%	N.A.	311.7%	364.0%	453.9%	486.3%	470.5%	249.2%	161.4%	120.2%	-96%	920%	431%	188%	72%	62%
Operating income	-104.2%	-276.9%	396.4%	-346.5%	3986.2%	758.8%	763.1%	693.2%	600.4%	273.8%	168.9%	124.5%	-120%	-257%	782%	202%	73%	62%
Net income	-236.1%	-258.0%	1130.9%	572.7%	530.8%	-1569.9%	495.5%	407.4%	486.2%	232.7%	160.1%	125.0%	-96%	856%	644%	189%	70%	63%

Source: Company data, Goldman Sachs Global Investment Research

Kinsus

We are Buy rated with a new 12-month TP of NT\$1,120 (from NT\$555), based on the same 16.5x P/E on 2028E EPS (rolled over from 2027E). We believe the company could trade at 1.2x STDV above the average PE multiple during previous upcycle of 14.2x as we expect the company's profitability to continue improving driven by increasing AI exposure and better ABF substrate pricing conditions, and factor in our latest earnings revisions.

Exhibit 30: Kinsus P&L table

NT\$m	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025E	2026E	2027E	2028E	2029E
Revenue	8,620	9,564	10,356	10,811	11,105	12,496	14,756	16,592	17,988	22,768	26,835	28,054	30,535	39,351	54,949	95,645	154,185	236,684
Gross profit	1,942	2,016	1,964	2,393	2,349	3,282	4,533	5,365	6,037	7,695	9,290	10,116	8,668	8,314	15,529	33,138	57,734	94,697
Operating expense	(1,440)	(1,396)	(1,336)	(1,473)	(1,459)	(1,565)	(1,719)	(1,891)	(2,065)	(2,459)	(2,979)	(3,142)	(7,572)	(5,645)	(6,634)	(10,645)	(17,311)	(26,486)
Operating income	501	620	628	921	890	1,717	2,814	3,473	3,972	5,236	6,311	6,974	1,095	2,670	8,895	22,494	40,422	68,211
Pretax income	700	619	713	1,055	964	1,817	2,914	3,473	4,022	5,286	6,311	6,959	1,603	3,087	9,168	22,579	40,507	68,396
Taxes expense	(88)	(65)	(100)	(118)	(89)	(309)	(495)	(590)	(724)	(952)	(1,136)	(1,253)	(272)	(370)	(1,483)	(4,064)	(7,291)	(12,311)
Net income	276	338	339	642	549	1,093	2,065	2,543	2,978	3,920	4,793	5,339	49	1,596	6,250	17,030	31,701	54,488
EPS, NT\$	0.61	0.74	0.75	1.41	1.17	2.34	4.42	5.44	6.37	8.38	10.25	11.42	0.11	3.52	13.36	36.42	67.79	116.52

Ratio analysis and assumption																		
As % of sales																		
Gross margin	22.5%	21.1%	19.0%	22.1%	21.2%	26.3%	30.7%	32.3%	33.6%	33.8%	34.6%	36.1%	28.4%	21.1%	28.3%	34.6%	37.4%	40.0%
Operating expense ratio	16.7%	14.6%	12.9%	13.6%	13.1%	12.5%	11.7%	11.4%	11.5%	10.8%	11.1%	11.2%	24.8%	14.3%	12.1%	11.1%	11.2%	11.2%
Operating margin	5.8%	6.5%	6.1%	8.5%	8.0%	13.7%	19.1%	20.9%	22.1%	23.0%	23.5%	24.9%	3.6%	6.8%	16.2%	23.5%	26.2%	28.8%
Net margin	3.2%	3.5%	3.3%	5.9%	4.9%	8.7%	14.0%	15.3%	16.6%	17.2%	17.9%	19.0%	0.2%	4.1%	11.4%	17.8%	20.6%	23.0%
QoQ growth (%)																		
Revenue	7.2%	11.0%	8.3%	4.4%	2.7%	12.5%	18.1%	12.4%	8.4%	26.6%	17.9%	4.5%	-	-	-	-	-	-
Gross profit	-13.6%	3.8%	-2.6%	21.9%	-1.8%	39.7%	38.1%	18.4%	12.5%	27.5%	20.7%	8.9%	-	-	-	-	-	-
Operating income	NM	23.7%	1.3%	46.6%	-3.3%	92.8%	63.9%	23.4%	14.4%	31.8%	20.5%	10.5%	-	-	-	-	-	-
Net income	-210.7%	22.5%	0.3%	89.3%	-14.5%	99.1%	88.9%	23.2%	17.1%	31.6%	22.3%	11.4%	-	-	-	-	-	-
Revenue	23.2%	31.0%	26.3%	34.4%	28.8%	30.7%	42.5%	53.5%	62.0%	82.2%	81.9%	69.1%	14%	29%	40%	74%	61%	54%
Gross profit	2.7%	-5.1%	-18.4%	6.5%	21.0%	62.8%	130.8%	124.2%	157.0%	134.5%	104.9%	88.6%	28%	-4%	87%	113%	74%	64%
Operating income	56.0%	74.5%	48.8%	NM	77.7%	176.9%	348.1%	277.2%	346.1%	205.0%	124.3%	100.8%	5%	144%	233%	153%	80%	69%
Net income	1026.8%	281.9%	83.2%	-357.5%	98.9%	223.2%	508.5%	295.9%	442.5%	258.6%	132.1%	110.0%	3%	3164%	292%	173%	86%	72%

Source: Company data, Goldman Sachs Global Investment Research

Unimicron

We are Neutral rated with a new 12-month TP of NT\$1,140 (from NT\$630), based on the same 17.4x P/E on 2028E EPS (rolled over from 2027E). We believe the company could trade at 2.7x STDV higher than past upcycle average P/E of 10.3x, and factor in our latest earnings revisions.

Exhibit 31: Unimicron P&L table

NT\$m	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026E	2027E	2028E	2029E
Revenue	30,090	32,466	33,994	34,691	37,446	42,920	50,138	58,012	60,983	67,742	79,801	90,648	115,373	131,241	188,516	299,174	424,900	583,347
Gross profit	4,026	4,246	4,549	5,472	6,726	9,306	12,435	15,866	18,299	21,819	27,337	33,819	16,315	18,292	44,333	101,274	166,786	246,555
Operating expense	(2,759)	(2,736)	(3,016)	(3,106)	(3,969)	(3,562)	(3,961)	(4,467)	(4,540)	(4,955)	(5,843)	(6,649)	(11,198)	(11,618)	(15,960)	(21,987)	(31,076)	(39,941)
Operating income	1,267	1,509	1,533	2,366	2,757	5,743	8,474	11,399	13,759	16,864	21,493	27,170	5,117	6,674	28,373	79,287	135,710	206,613
Pretax income	1,373	357	2,947	4,150	6,299	7,293	8,474	11,399	13,759	16,864	21,493	27,170	7,321	8,827	33,465	79,287	133,710	206,613
Taxes expense	(450)	(97)	(496)	(235)	(918)	(1,605)	(1,949)	(2,215)	(2,752)	(3,710)	(4,944)	(5,281)	(1,765)	(1,277)	(6,687)	(16,686)	(29,768)	(43,676)
Net income	915	30	2,194	3,535	5,043	5,339	6,225	8,884	10,707	12,854	16,250	21,589	5,082	6,673	25,490	61,401	102,742	162,368
EPS, NT\$	0.60	0.02	1.44	2.32	3.28	3.47	4.05	5.78	7.03	8.44	10.67	14.18	3.34	4.38	16.58	40.33	66.82	106.66

Ratio analysis and assumption																		
As % of sales																		
Gross margin	13.4%	13.1%	13.4%	15.8%	18.0%	21.7%	24.8%	27.3%	30.0%	32.2%	34.3%	37.3%	14.1%	13.9%	23.5%	33.9%	39.3%	42.3%
Operating expense ratio	9.2%	8.4%	8.9%	9.0%	10.6%	8.3%	7.9%	7.7%	7.4%	7.3%	7.3%	7.3%	9.7%	8.9%	8.5%	7.3%	7.3%	6.8%
Operating margin	4.2%	4.6%	4.5%	6.8%	7.4%	13.4%	16.9%	19.6%	22.6%	24.9%	26.9%	30.0%	4.4%	5.1%	15.1%	26.5%	31.9%	35.4%
Net margin	3.0%	0.1%	6.5%	10.2%	13.5%	12.4%	15.3%	17.6%	19.0%	20.4%	23.8%		4.4%	5.1%	13.5%	20.5%	24.2%	27.8%
QoQ growth (%)																		
Revenue	2.4%	7.9%	4.7%	2.0%	7.9%	14.6%	16.8%	15.7%	75.8%	11.1%	17.8%	13.6%	-	-	-	-	-	-
Gross profit	18.5%	5.5%	7.1%	20.3%	22.9%	38.4%	33.6%	27.6%	234.4%	19.2%	25.3%	23.7%	-	-	-	-	-	-
Operating income	86.5%	19.1%	1.5%	54.4%	16.5%	108.3%	47.5%	34.5%	481.6%	22.6%	27.4%	26.4%	-	-	-	-	-	-
Net income	1531.2%	-96.8%	7312.1%	61.1%	42.7%	5.9%	16.6%	42.7%	202.9%	20.1%	26.4%	32.9%	-	-	-	-	-	-
YoY growth (%)																		
Revenue	14.0%	16.5%	7.2%	18.1%	24.4%	32.2%	47.5%	67.2%	102.7%	108.7%	134.7%	161.3%	11%	14%	44%	59%	42%	37%
Gross profit	-6.2%	15.3%	-8.0%	61.0%	67.1%	119.2%	173.4%	189.9%	354.5%	413.9%	501.0%	518.0%	-20%	12%	142%	128%	65%	48%
Operating income	-19.7%	68.3%	-21.9%	248.2%	117.6%	280.5%	452.9%	381.9%	986.1%	1017.3%	1302.5%	1048.5%	-43%	30%	325%	179%	71%	52%
Net income	-62.4%	-98.1%	120.1%	6204.2%	451.4%	N/A	183.7%	151.3%	1070.8%	N/A	640.5%	510.8%	-58%	31%	282%	141%	67%	58%

Source: Company data, Goldman Sachs Global Investment Research

ZDT

We are Buy rated with a new 12-month TP NT\$824 (from NT\$388), still based on our SOTP based valuation methodology of (1) 3.5x blended P/B (was 3.0x blended P/B) for ABF & BT substrate, which is 1 STDV above early upcycle industry average, (2) 20x P/E for high-growth AI server/switch related PCB (in line with industry average), and (3) 8x P/E for low-growth consumer electronics PCB (in-line with industry downcycle average valuation), and roll over our valuation period from 4Q26-3Q27 to 2028E, and factor in our latest earnings revision.

Exhibit 32: ZDT P&L table

NT\$m	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26E	3Q26E	4Q26E	1Q27E	2Q27E	3Q27E	4Q27E	2024	2025	2026E	2027E	2028E	2029E
Revenue	40,082	38,203	47,366	56,870	40,728	48,431	63,831	71,690	64,038	70,652	89,912	103,161	171,664	182,522	224,680	327,764	454,047	602,605
Gross profit	5,885	7,011	10,410	12,830	8,812	10,430	15,637	18,261	13,929	16,830	25,231	29,029	32,461	36,136	53,140	85,019	133,376	197,093
Operating expense	(4,829)	(4,586)	(5,880)	(6,909)	(6,308)	(6,417)	(7,149)	(7,384)	(6,480)	(7,542)	(8,714)	(9,077)	(20,875)	(22,205)	(27,258)	(31,813)	(43,392)	(57,767)
Operating income	1,056	2,425	4,530	5,921	2,503	4,013	8,488	10,877	7,449	9,288	16,517	19,951	11,586	13,932	25,882	53,206	89,985	139,326
Pretax income	1,457	2,272	4,661	5,673	2,400	4,113	8,988	11,877	8,249	9,988	16,617	20,951	15,045	14,063	27,378	55,806	91,285	140,626
Taxes expense	(431)	(885)	(1,072)	(1,070)	(353)	(699)	(1,348)	(1,782)	(1,402)	(1,698)	(2,493)	(3,143)	(1,948)	(3,458)	(4,182)	(8,736)	(14,310)	(22,308)
Net income	632	605	2,392	3,161	1,426	2,418	5,393	7,126	4,908	5,873	9,970	12,571	9,180	6,791	16,363	33,322	54,498	83,856
EPS, NT\$	0.66	0.63	2.46	3.16	1.33	2.26	5.03	6.65	4.58	5.48	9.30	11.73	9.67	6.91	15.27	31.09	50.84	78.23

Ratio analysis and assumption																		
As % of sales																		
Gross margin	14.7%	18.4%	22.0%	22.6%	21.6%	21.5%	24.5%	25.5%	21.8%	23.8%	28.1%	28.1%	18.9%	19.8%	23.7%	25.9%	29.4%	32.7%
Operating expense ratio	12.0%	12.0%	12.4%	12.1%	15.5%	13.3%	11.2%	10.3%	10.1%	10.7%	9.7%	8.8%	12.2%	12.2%	12.1%	9.7%	9.6%	9.6%
Operating margin	2.6%	6.3%	9.6%	10.4%	6.1%	8.3%	13.3%	15.2%	11.6%	13.1%	18.4%	19.3%	6.7%	7.6%	11.5%	16.2%	19.8%	23.1%
Net margin	1.6%	1.6%	5.0%	5.6%	3.5%	5.0%	8.4%	9.9%	7.7%	8.3%	11.1%	12.2%	5.3%	3.7%	7.3%	10.2%	12.0%	13.9%
QoQ growth (%)																		
Revenue	-28.6%	-4.7%	24.0%	20.1%	-28.4%	18.9%	31.8%	12.3%	-10.7%	10.3%	27.3%	14.7%	-	-	-	-	-	-
Gross profit	-48.7%	19.1%	48.5%	23.2%	-31.3%	18.4%	49.9%	16.8%	-23.7%	20.8%	49.9%	15.1%	-	-	-	-	-	-
Operating income	-81.0%	129.6%	86.8%	30.7%	-57.7%	60.3%	111.5%	28.1%	-31.5%	24.7%	77.8%	20.8%	-	-	-	-	-	-
Net income	-85.5%	-4.3%	295.3%	32.2%	-54.9%	69.6%	123.0%	32.1%	-31.1%	19.7%	69.8%	26.1%	-	-	-	-	-	-
YoY growth (%)																		
Revenue	23.3%	17.9%	-6.4%	1.3%	1.6%	26.8%	34.8%	26.1%	57.2%	45.9%	40.9%	43.9%	13%	6%	23%	46%	39%	33%
Gross profit	10.3%	65.1%	-8.7%	11.9%	49.7%	48.8%	50.2%	42.3%	58.1%	61.4%	61.4%	59.0%	18%	11%	47%	60%	57%	48%
Operating income	42.2%	-476.8%	-23.6%	6.6%	137.1%	65.5%	87.4%	83.7%	197.6%	131.5%	94.6%	83.4%	26%	20%	86%	106%	69%	55%
Net income	-35.3%	24.5%	-28.7%	-27.5%	125.4%	299.7%	125.5%	125.4%	244.3%	142.8%	84.9%	76.4%	48%	-26%	141%	104%	64%	54%

Source: Company data, Goldman Sachs Global Investment Research

Investment thesis, PT methodology and risks

Investment Thesis - NYPCB

NYPCB manufactures and markets printed circuit boards (PCBs) and integrated circuit (IC) substrates (including ABF and BT substrates). We are Buy-rated on NYPCB considering the company's high exposure to BT substrates and non-LTA ABF substrate (70%+ of total revenue), where we expect GM/OPM to increase significantly in the coming quarters (from breakeven in OPM in 1H25 to 27/41% in 2026/27E) due to the favorable substrate pricing outlook. Despite the company's low exposure to high-end ABF substrate industry (<20% of total revenue in 1H25), we believe the expanding demand from ASIC AI server and high-end switch IC demand will drive the company's high-end ABF revenue exposure to 40%+ by 2027E, suggesting better GM/OPM outlook in the long term. In 2H25-2H26, we believe NYPCB will continue to benefit from stronger ABF/BT pricing and should be a key beneficiary of the pricing uptrend considering the company's key focus on BT and non-LTA ABF substrate industries. Moreover, in the long term (after 2026), we expect the ABF industry to return to supply tightness, and NYPCB's expanding high-end exposure will lead to a long term earnings growth outlook (122% 2026-28E earnings CAGR). As a result, we see solid upside potential for NYPCB's earnings and stock performance.

Price Target Risks and Methodology - NYPCB

Valuation methodology: Our 12-month target price of NT\$2,310 for NYPCB is based on 21x 2028 P/E, which is +1.5x STDV higher than the past upcycle average P/E (13.3x).

Key downside risks: (1) Slower-than-expected PC demand recovery, (2) Slower-than-expected ABF substrates pricing upgrading outlook, and (3) Slower-than-expected NYPCB new high-end capacity qualification progress.

Investment Thesis - Kinsus

Kinsus is one of the key IC substrates suppliers globally with ~10% ABF/BT substrates market share in 2024, with ~18% of total 2025 revenue contributed by its contact lens subsidiary, Pegavision (6941.TW, Not Covered). With our view that the ABF substrate shortage started from 2Q26 (ABF substrate return to supply demand balance from late 2025), we believe Kinsus' performance will continue to improve compared with high-end ABF peers considering the company's increasing market share in the high-end ABF substrate market; however, ~36% of the company's revenue was contributed by BT substrates in 2025 and 30% of the company's ABF revenue is from non-LTA customers, which we believe will continue to enjoy a better pricing outlook. As a result, we believe Kinsus' earnings outlook should continue to be strong (~125% 2026-28E CAGR), mainly driven by better BT substrate and non-LTA ABF substrate pricing outlook, despite the unfavorable demand outlook before 2H26. We believe Kinsus is an experienced substrate supplier with a clear capacity expansion plan that can also upgrade its product portfolio in the long term. We believe we are at the beginning of an industry upcycle, making us positive on Kinsus' stock performance over the next 12 months.

Price Target Risks and Methodology - Kinsus

Valuation methodology: Our 12-month target price of NT\$1,120 for Kinsus is based on

16.5x 2028E P/E, which is +1.2x STDV higher than Kinsus' past upcycle average P/E (14.2x).

Key downside risks: (1) Slower-than-expected PC demand recovery, (2) Slower-than-expected ABF substrates pricing upgrading outlook, (3) Slower/Less-than-expected AI server orders from 2H26.

Investment Thesis - Unimicron Technology

Unimicron is a key supplier of ABF substrates (25% market share in 2025) and BT substrates and also produces FPC/HDI/PCB products. We are Neutral rated on the shares given long-term growth opportunity prospects: (1) over 70% of ABF shipments are covered by long-term agreements (LTA), which limits pricing flexibility and makes price hike less likely; (2) the company's execution track record has been weaker than peers; (3) capacity expansion plan remains slower than peers, constraining its ability to capture incremental AI demand. The current ABF substrates market is seeing more favorable supply demand outlook, and Unimicron is likely to see better pricing outlook to reflect rising material costs in the coming quarters. AI ABF account for 10% of the company's total revenue, but the growth is likely to be slower than peers due to the company continuing to lose market share to peers given its less favourable production yield and slow capacity expansion. Overall, we are Neutral rated on Unimicron given our outlook for the limited upside potential, barring ASIC AI servers, despite our generally positive view on the long term on profitability trajectory.

Price Target Risks and Methodology - Unimicron Technology

Valuation methodology: Our 12-month target price of NT\$1,140 for Unimicron is based on 17.4x 2028E P/E, which is 2.7x STDV higher than the past upcycle average P/E (10.3x).

Key upside/downside risks: (1) Slower/faster-than-expected PC demand recovery, (2) Slower/faster-than-expected ABF substrates upgrading pace, and (3) Slower/faster than-expected AI server PCB market share loss progress.

Zhen Ding Technology (ZDT) is the global leading PCB supplier (top 1 in market share in sales) with 7-10% market share, and started its ABF substrate business from 2023 in Shenzhen, China, and has the highest ABF revenue exposure to the China market compared to all other ABF substrate suppliers. We believe ZDT will hold more than 40% ABF substrate market share in China by 2027E (up from 31% in 2024) and benefit the most from the strong and growing China ABF substrate market (53% 2025E-27E CAGR) due to the increasing in-sourcing ratio for the AI computing ASIC/GPU market in China. We also expect the increasing foldable smartphone penetration rate in the long term to drive ZDT's FPC TAM, and account for 50% of the company's total revenue in 2027E. We see further revenue upside potential from foldable phones. We are Buy rated mainly due to our positive view on the company's China ABF substrate business opportunity, which should not only drive ZDT's revenue growth, but also improve its OPM level in the long term. We see upside for the stock price considering the company's solid growth opportunity, and our TP implied P/E multiple is still lower than the company's past 10+ year P/E average, which we believe is fair considering the potential upside from China ABF substrate market and increasing foldable phone penetration rate.

Valuation: Our 12-month TP of NT\$824 is based on a SOTP valuation methodology,

including (1) 3.5x blended 2028E P/B for ABF & BT substrate, (2) 20x 2028E P/E for high-growth AI server/switch related PCB (in line with industry average), and (3) 8x 2028E P/E for low-growth consumer electronics PCB (in-line with industry downcycle average valuation).

Key downside risks include: (1) weaker-than-expected iPhone demand or slower content upgrade, (2) delay in new substrate capacity or slower-than-expected yield rate ramp up progress, and (3) fiercer competition in China ABF market.

Disclosure Appendix

Reg AC

We, Chao Wang, Daiki Takayama, Allen Chang and Al Wang, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

Contributing Authors: Chao Wang Goldman Sachs (Asia) L.L.C., Taipei Branch, Daiki Takayama Goldman Sachs Japan Co., Ltd., Allen Chang Goldman Sachs (Asia) L.L.C., Al Wang Goldman Sachs (Asia) L.L.C., Taipei Branch.

Unless otherwise stated, the individuals listed in the Contributing Authors disclosure of this report are analysts in Goldman Sachs' Global Investment Research division.

GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

Quantum is Goldman Sachs' proprietary database providing access to detailed financial statement histories, forecasts and ratios. It can be used for in-depth analysis of a single company, or to make comparisons between companies in different sectors and markets.

Disclosures

The rating(s) for Kinsus, NYPCB, Unimicron Technology and Zhen Ding Technology Holding is/are relative to the other companies in its/their coverage universe: AWSC, Airtac International Group, Co-Tech Development Corp., Delta Electronics, Elite Material, GCE, Hiwin Corp., ITEQ Corp, Kinsus, Lotes, NYPCB, Taiwan Union Technology Corp., Unimicron Technology, Win Semiconductors Corp., Yageo Corp., Zhen Ding Technology Holding

Company-specific regulatory disclosures

The following disclosures relate to relationships between The Goldman Sachs Group, Inc. (with its affiliates, "Goldman Sachs") and companies covered by Goldman Sachs Global Investment Research and referred to in this research.

Goldman Sachs beneficially owned 1% or more of common equity (excluding positions managed by affiliates and business units not required to be aggregated under US securities law) as of the month end preceding this report: Kinsus (NT\$795.00) and NYPCB (NT\$1,215.00)

Goldman Sachs expects to receive or intends to seek compensation for investment banking services in the next 3 months: NYPCB (NT\$1,215.00) and Zhen Ding Technology Holding (NT\$588.00)

Goldman Sachs had an investment banking services client relationship during the past 12 months with: NYPCB (NT\$1,215.00) and Zhen Ding Technology Holding (NT\$588.00)

Goldman Sachs had a non-securities services client relationship during the past 12 months with: NYPCB (NT\$1,215.00)

There are no company-specific disclosures for: Unimicron Technology (NT\$875.00)

Distribution of ratings/investment banking relationships

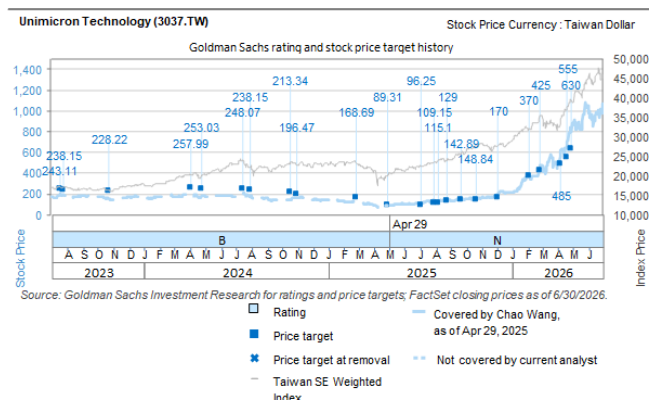
Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution				Investment Banking Relationships		
	Buy	Hold	Sell		Buy	Hold	Sell
Global	50%	34%	16%		65%	59%	43%

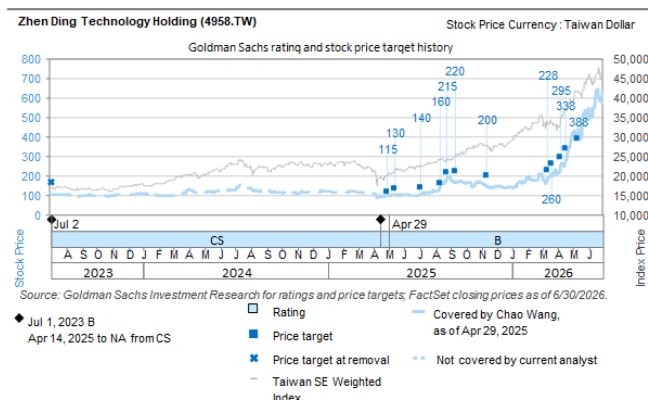
As of July 1, 2026, Goldman Sachs Global Investment Research had investment ratings on 3,104 equity securities. Goldman Sachs assigns stocks as Buys and Sells on various regional Investment Lists; stocks not so assigned are deemed Neutral. Such assignments equate to Buy, Hold and Sell for the

purposes of the above disclosure required by the FINRA Rules. See 'Ratings, Coverage universe and related definitions' below. The Investment Banking Relationships chart reflects the percentage of subject companies within each rating category for whom Goldman Sachs has provided investment banking services within the previous twelve months.

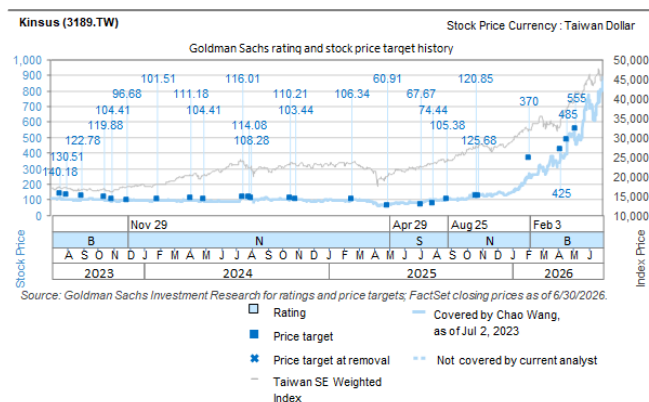
Price target and rating history chart(s)



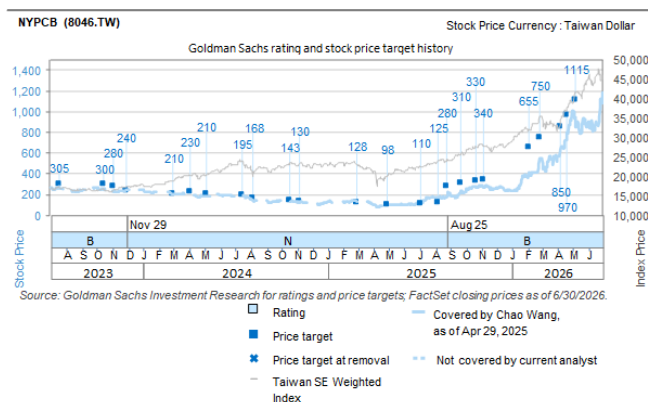
The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.



The price targets shown should be considered in the context of all prior published Goldman Sachs research, which may or may not have included price targets, as well as developments relating to the company, its industry and financial markets.

Target price history table(s)

Zhen Ding Technology Holding (4958.TW)

Date of report	Target price (NT\$)	Closing price (NT\$)
12-May-26	388.00	407.00
17-Apr-26	338.00	282.00
08-Apr-26	295.00	247.50
20-Mar-26	260.00	209.50
13-Mar-26	228.00	188.50
12-Nov-25	200.00	144.50
10-Sep-25	220.00	168.50
25-Aug-25	215.00	180.50
12-Aug-25	160.00	147.00
04-Jul-25	140.00	103.00
13-May-25	130.00	104.50
29-Apr-25	115.00	97.10

Kinsus (3189.TW)

Date of report	Target price (NT\$)	Closing price (NT\$)
08-May-26	555.00	458.00
21-Apr-26	485.00	473.50
08-Apr-26	425.00	394.00
03-Feb-26	370.00	246.00
27-Oct-25	130.00	143.57
21-Oct-25	125.00	145.02
25-Aug-25	109.00	103.93
29-Jul-25	77.00	102.48
04-Jul-25	70.00	85.27
29-Apr-25	63.00	73.57
17-Feb-25	110.00	96.68
28-Oct-24	107.00	108.76
18-Oct-24	114.00	101.51
01-Aug-24	112.00	107.31
29-Jul-24	118.00	115.05
16-Jul-24	120.00	116.01
29-Apr-24	108.00	94.65
03-Apr-24	115.00	98.61
29-Jan-24	105.00	98.13

Zhen Ding Technology Holding (4958.TW)

Date of report	Target price (NT\$)	Closing price (NT\$)
----------------	---------------------	----------------------

Kinsus (3189.TW)

Date of report	Target price (NT\$)	Closing price (NT\$)
----------------	---------------------	----------------------

29-Nov-23	100.00	96.00
31-Oct-23	108.00	91.65
16-Oct-23	124.00	98.13
01-Sep-23	127.00	103.44
01-Aug-23	135.00	103.44
18-Jul-23	145.00	116.50

Unimicron Technology (3037.TW)

Date of report	Target price (NT\$)	Closing price (NT\$)
----------------	---------------------	----------------------

28-Apr-26	630.00	825.00
21-Apr-26	555.00	710.00
08-Apr-26	485.00	620.00
25-Feb-26	425.00	458.50
03-Feb-26	370.00	385.00
04-Dec-25	170.00	198.50
21-Oct-25	150.00	158.76
22-Sep-25	144.00	157.28
25-Aug-25	130.00	138.92
07-Aug-25	116.00	134.95
30-Jul-25	110.00	140.90
04-Jul-25	97.00	117.09
29-Apr-25	90.00	95.26
26-Feb-25	170.00	124.53
30-Oct-24	198.00	160.25
18-Oct-24	215.00	143.38
31-Jul-24	240.00	178.11
16-Jul-24	250.00	196.47
24-Apr-24	255.00	185.56
03-Apr-24	260.00	190.52
24-Oct-23	230.00	155.79
26-Jul-23	240.00	195.97
18-Jul-23	245.00	198.46

NYPCB (8046.TW)

Date of report	Target price (NT\$)	Closing price (NT\$)
----------------	---------------------	----------------------

07-May-26	1,115.00	947.00
21-Apr-26	970.00	789.00
08-Apr-26	850.00	637.00
25-Feb-26	750.00	553.00
03-Feb-26	655.00	394.50
05-Nov-25	340.00	272.00
21-Oct-25	330.00	273.50
22-Sep-25	310.00	235.50
25-Aug-25	280.00	186.00
07-Aug-25	125.00	172.50
04-Jul-25	110.00	126.00
29-Apr-25	98.00	101.00
28-Feb-25	128.00	136.00
06-Nov-24	130.00	124.50
18-Oct-24	143.00	133.50
07-Aug-24	168.00	143.00
16-Jul-24	195.00	183.50
06-May-24	210.00	185.00
03-Apr-24	230.00	209.00
01-Mar-24	210.00	210.50
29-Nov-23	240.00	251.00
03-Nov-23	280.00	232.50
16-Oct-23	300.00	253.00
18-Jul-23	305.00	275.50

Price targets shown in table(s) are unadjusted for corporate actions.

Regulatory disclosures**Disclosures required by United States laws and regulations**

See company-specific regulatory disclosures above for any of the following disclosures required as to companies referred to in this report: manager or co-manager in a pending transaction; 1% or other ownership; compensation for certain services; types of client relationships; managed/co-managed public offerings in prior periods; directorships; for equity securities, market making and/or specialist role. Goldman Sachs trades or may trade as a principal in debt securities (or in related derivatives) of issuers discussed in this report.

The following are additional required disclosures: **Ownership and material conflicts of interest:** Goldman Sachs policy prohibits its analysts, professionals reporting to analysts and members of their households from owning securities of any company in the analyst's area of coverage. **Analyst compensation:** Analysts are paid in part based on the profitability of Goldman Sachs, which includes investment banking revenues. **Analyst as officer or director:** Goldman Sachs policy generally prohibits its analysts, persons reporting to analysts or members of their households from serving as an officer, director or advisor of any company in the analyst's area of coverage. **Non-U.S. Analysts:** Non-U.S. analysts may not be associated persons of Goldman Sachs & Co. LLC and therefore may not be subject to FINRA Rule 2241 or FINRA Rule 2242 restrictions on communications with a subject company, public appearances and trading in securities covered by the analysts.

Additional disclosures required under the laws and regulations of jurisdictions other than the United States

The following disclosures are those required by the jurisdiction indicated, except to the extent already made above pursuant to United States laws and regulations. **Australia:** Goldman Sachs Australia Pty Ltd and its affiliates are not authorised deposit-taking institutions (as that term is defined in the Banking Act 1959 (Cth)) in Australia and do not provide banking services, nor carry on a banking business, in Australia. This research, and any access to it, is intended only for "wholesale clients" within the meaning of the Australian Corporations Act, unless otherwise agreed by Goldman Sachs. In producing research reports, members of Global Investment Research of Goldman Sachs Australia may attend site visits and other meetings hosted by the companies and other entities which are the subject of its research reports. In some instances the costs of such site visits or meetings may be met in part or in whole by the issuers concerned if Goldman Sachs Australia considers it is appropriate and reasonable in the specific circumstances relating to the site visit or meeting. To the extent that the contents of this document contains any financial product advice, it is general advice only and has been prepared by Goldman Sachs without taking into account a client's objectives, financial situation or needs. A client should, before acting on any such advice, consider the appropriateness of the advice having regard to the client's own objectives, financial situation and needs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests and a copy of Goldman Sachs' Australian Sell-Side Research Independence Policy Statement are available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Brazil:** Disclosure information in relation to CVM Resolution n. 20 is available at <https://www.gs.com/worldwide/brazil/area/gir/index.html>. Where applicable, the Brazil-registered analyst

primarily responsible for the content of this research report, as defined in Article 20 of CVM Resolution n. 20, is the first author named at the beginning of this report, unless indicated otherwise at the end of the text. **Canada:** This information is being provided to you for information purposes only and is not, and under no circumstances should be construed as, an advertisement, offering or solicitation by Goldman Sachs & Co. LLC for purchasers of securities in Canada to trade in any Canadian security. Goldman Sachs & Co. LLC is not registered as a dealer in any jurisdiction in Canada under applicable Canadian securities laws and generally is not permitted to trade in Canadian securities and may be prohibited from selling certain securities and products in certain jurisdictions in Canada. If you wish to trade in any Canadian securities or other products in Canada please contact Goldman Sachs Canada Inc., an affiliate of The Goldman Sachs Group Inc., or another registered Canadian dealer. **Hong Kong:** Further information on the securities of covered companies referred to in this research may be obtained on request from Goldman Sachs (Asia) L.L.C. **India:** Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (India) Securities Private Limited, Research Analyst – SEBI Registration Number INH000001493, 10th Floor, Ascent-Worli, Sudam Kalu Ahire Marg, Worli, Mumbai-400 025, India, Corporate Identity Number U74140MH2006FTC160634, Phone +91 22 6616 9000, Fax +91 22 6616 9001. Goldman Sachs may beneficially own 1% or more of the securities (as such term is defined in clause 2 (h) the Indian Securities Contracts (Regulation) Act, 1956) of the subject company or companies referred to in this research report. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Goldman Sachs (India) Securities Private Limited compliance officer and investor grievance contact details, a copy of the annual compliance audit report and other relevant information and disclosures can be found at this link: <https://www.goldmansachs.com/worldwide/india/research-analyst>. **Japan:** See below. **Korea:** This research, and any access to it, is intended only for “professional investors” within the meaning of the Financial Services and Capital Markets Act, unless otherwise agreed by Goldman Sachs. Further information on the subject company or companies referred to in this research may be obtained from Goldman Sachs (Asia) L.L.C., Seoul Branch. **New Zealand:** Goldman Sachs New Zealand Limited and its affiliates are neither “registered banks” nor “deposit takers” (as defined in the Reserve Bank of New Zealand Act 1989) in New Zealand. This research, and any access to it, is intended for “wholesale clients” (as defined in the Financial Advisers Act 2008) unless otherwise agreed by Goldman Sachs. A copy of certain Goldman Sachs Australia and New Zealand disclosure of interests is available at: <https://www.goldmansachs.com/disclosures/australia-new-zealand/index.html>. **Russia:** Research reports distributed in the Russian Federation are not advertising as defined in the Russian legislation, but are information and analysis not having product promotion as their main purpose and do not provide appraisal within the meaning of the Russian legislation on appraisal activity. Research reports do not constitute a personalized investment recommendation as defined in Russian laws and regulations, are not addressed to a specific client, and are prepared without analyzing the financial circumstances, investment profiles or risk profiles of clients. Goldman Sachs assumes no responsibility for any investment decisions that may be taken by a client or any other person based on this research report. **Singapore:** Goldman Sachs (Singapore) Pte. (Company Number: 198602165W), which is regulated by the Monetary Authority of Singapore, accepts legal responsibility for this research, and should be contacted with respect to any matters arising from, or in connection with, this research. **Taiwan:** This material is for reference only and must not be reprinted without permission. Investors should carefully consider their own investment risk. Investment results are the responsibility of the individual investor. **United Kingdom:** Persons who would be categorized as retail clients in the United Kingdom, as such term is defined in the rules of the Financial Conduct Authority, should read this research in conjunction with prior Goldman Sachs research on the covered companies referred to herein and should refer to the risk warnings that have been sent to them by Goldman Sachs International. A copy of these risks warnings, and a glossary of certain financial terms used in this report, are available from Goldman Sachs International on request.

European Union and United Kingdom: Disclosure information in relation to Article 6 (2) of the European Commission Delegated Regulation (EU) (2016/958) supplementing Regulation (EU) No 596/2014 of the European Parliament and of the Council (including as that Delegated Regulation is implemented into United Kingdom domestic law and regulation following the United Kingdom’s departure from the European Union and the European Economic Area) with regard to regulatory technical standards for the technical arrangements for objective presentation of investment recommendations or other information recommending or suggesting an investment strategy and for disclosure of particular interests or indications of conflicts of interest is available at <https://www.gs.com/disclosures/europeanpolicy.html> which states the European Policy for Managing Conflicts of Interest in Connection with Investment Research.

Japan: Goldman Sachs Japan Co., Ltd. is a Financial Instrument Dealer registered with the Kanto Financial Bureau under registration number Kinsho 69, and a member of Japan Securities Dealers Association, Financial Futures Association of Japan Type II Financial Instruments Firms Association, and Investment Management Association of Japan. Sales and purchase of equities are subject to commission pre-determined with clients plus consumption tax. See company-specific disclosures as to any applicable disclosures required by Japanese stock exchanges, the Japanese Securities Dealers Association or the Japanese Securities Finance Company.

Ratings, coverage universe and related definitions

Buy (B), Neutral (N), Sell (S) Analysts recommend stocks as Buys or Sells for inclusion on various regional Investment Lists. Being assigned a Buy or Sell on an Investment List is determined by a stock’s total return potential relative to its coverage universe. Any stock not assigned as a Buy or a Sell on an Investment List with an active rating (i.e., a stock that is not Rating Suspended, Not Rated, Early-Stage Biotech, Coverage Suspended or Not Covered), is deemed Neutral. Each region manages Regional Conviction Lists, which are selected from Buy rated stocks on the respective region’s Investment Lists and represent investment recommendations focused on the size of the total return potential and/or the likelihood of the realization of the return across their respective areas of coverage. The addition or removal of stocks from such Conviction Lists are managed by the Investment Review Committee or other designated committee in each respective region and do not represent a change in the analysts’ investment rating for such stocks.

Total return potential represents the upside or downside differential between the current share price and the price target, including all paid or anticipated dividends, expected during the time horizon associated with the price target. Price targets are required for all covered stocks. The total return potential, price target and associated time horizon are stated in each report adding or reiterating an Investment List membership.

Coverage Universe: A list of all stocks in each coverage universe is available by primary analyst, stock and coverage universe at <https://www.gs.com/research/hedge.html>.

Not Rated (NR). The investment rating, target price and earnings estimates (where relevant) are removed pursuant to Goldman Sachs policy when Goldman Sachs is acting in an advisory capacity in a merger or in a strategic transaction involving this company, when there are legal, regulatory or policy constraints due to Goldman Sachs’ involvement in a transaction, and in certain other circumstances. **Early-Stage Biotech (ES).** An investment rating and a target price are not assigned pursuant to Goldman Sachs policy when this company has neither a drug, treatment or medical device that has passed a Phase II clinical trial nor a license to distribute a post-Phase II drug, treatment or medical device. **Rating Suspended (RS).** Goldman Sachs Research has suspended the investment rating and price target for this stock, because there is not a sufficient fundamental basis for determining an investment rating or target price. The previous investment rating and target price, if any, are no longer in effect for this stock and should not be relied upon. **Coverage Suspended (CS).** Goldman Sachs has suspended coverage of this company. **Not Covered (NC).** Goldman Sachs does not cover this company.

Global product; distributing entities

Goldman Sachs Global Investment Research produces and distributes research products for clients of Goldman Sachs on a global basis. Analysts based in Goldman Sachs offices around the world produce research on industries and companies, and research on macroeconomics, currencies, commodities and portfolio strategy. This research is disseminated in Australia by Goldman Sachs Australia Pty Ltd (ABN 21 006 797 897); in Brazil by Goldman Sachs do Brasil Corretora de Títulos e Valores Mobiliários S.A.; Public Communication Channel Goldman Sachs Brazil: 0800 727 5764 and / or contatogoldmanbrasil@gs.com. Available Weekdays (except holidays), from 9am to 6pm. Canal de Comunicação com o Público Goldman Sachs Brasil:

0800 727 5764 e/ou contatogoldmanbrasil@gs.com. Horário de funcionamento: segunda-feira à sexta-feira (exceto feriados), das 9h às 18h; in Canada by Goldman Sachs & Co. LLC; in Hong Kong by Goldman Sachs (Asia) L.L.C.; in India by Goldman Sachs (India) Securities Private Ltd.; in Japan by Goldman Sachs Japan Co., Ltd.; in the Republic of Korea by Goldman Sachs (Asia) L.L.C., Seoul Branch; in New Zealand by Goldman Sachs New Zealand Limited; in Russia by OOO Goldman Sachs; in Singapore by Goldman Sachs (Singapore) Pte. (Company Number: 198602165W); and in the United States of America by Goldman Sachs & Co. LLC. Goldman Sachs International has approved this research in connection with its distribution in the United Kingdom.

Goldman Sachs International ("GSI"), authorised by the Prudential Regulation Authority ("PRA") and regulated by the Financial Conduct Authority ("FCA") and the PRA, has approved this research in connection with its distribution in the United Kingdom.

European Economic Area: Goldman Sachs Bank Europe SE ("GSBE") is a credit institution incorporated in Germany and, within the Single Supervisory Mechanism, subject to direct prudential supervision by the European Central Bank and in other respects supervised by German Federal Financial Supervisory Authority (Bundesanstalt für Finanzdienstleistungsaufsicht, BaFin) and Deutsche Bundesbank and disseminates research within the European Economic Area.

General disclosures

This research is for our clients only. Other than disclosures relating to Goldman Sachs, this research is based on current public information that we consider reliable, but we do not represent it is accurate or complete, and it should not be relied on as such. The information, opinions, estimates and forecasts contained herein are as of the date hereof and are subject to change without prior notification. We seek to update our research as appropriate, but various regulations may prevent us from doing so. Other than certain industry reports published on a periodic basis, the large majority of reports are published at irregular intervals as appropriate in the analyst's judgment.

Goldman Sachs conducts a global full-service, integrated investment banking, investment management, and brokerage business. We have investment banking and other business relationships with a substantial percentage of the companies covered by Global Investment Research. Goldman Sachs & Co. LLC, the United States broker dealer, is a member of SIPC (<https://www.sipc.org>).

Our salespeople, traders, and other professionals may provide oral or written market commentary or trading strategies to our clients and principal trading desks that reflect opinions that are contrary to the opinions expressed in this research. Our asset management area, principal trading desks and investing businesses may make investment decisions that are inconsistent with the recommendations or views expressed in this research.

The analysts named in this report may have from time to time discussed with our clients, including Goldman Sachs salespersons and traders, or may discuss in this report, trading strategies that reference catalysts or events that may have a near-term impact on the market price of the equity securities discussed in this report, which impact may be directionally counter to the analyst's published price target expectations for such stocks. Any such trading strategies are distinct from and do not affect the analyst's fundamental equity rating for such stocks, which rating reflects a stock's return potential relative to its coverage universe as described herein.

We and our affiliates, officers, directors, and employees will from time to time have long or short positions in, act as principal in, and buy or sell, the securities or derivatives, if any, referred to in this research, unless otherwise prohibited by regulation or Goldman Sachs policy.

The views attributed to third party presenters at Goldman Sachs arranged conferences, including individuals from other parts of Goldman Sachs, do not necessarily reflect those of Global Investment Research and are not an official view of Goldman Sachs.

Any third party referenced herein, including any salespeople, traders and other professionals or members of their household, may have positions in the products mentioned that are inconsistent with the views expressed by analysts named in this report.

This research is not an offer to sell or the solicitation of an offer to buy any security in any jurisdiction where such an offer or solicitation would be illegal. It does not constitute a personal recommendation or take into account the particular investment objectives, financial situations, or needs of individual clients. Clients should consider whether any advice or recommendation in this research is suitable for their particular circumstances and, if appropriate, seek professional advice, including tax advice. The price and value of investments referred to in this research and the income from them may fluctuate. Past performance is not a guide to future performance, future returns are not guaranteed, and a loss of original capital may occur. Fluctuations in exchange rates could have adverse effects on the value or price of, or income derived from, certain investments.

Certain transactions, including those involving futures, options, and other derivatives, give rise to substantial risk and are not suitable for all investors. Investors should review current options and futures disclosure documents which are available from Goldman Sachs sales representatives or at <https://www.theocc.com/about/publications/character-risks.jsp> and https://www.goldmansachs.com/disclosures/cftc_fcm_disclosures. Transaction costs may be significant in option strategies calling for multiple purchase and sales of options such as spreads. Supporting documentation will be supplied upon request.

Differing Levels of Service provided by Global Investment Research: The level and types of services provided to you by Goldman Sachs Global Investment Research may vary as compared to that provided to internal and other external clients of GS, depending on various factors including your individual preferences as to the frequency and manner of receiving communication, your risk profile and investment focus and perspective (e.g., marketwide, sector specific, long term, short term), the size and scope of your overall client relationship with GS, and legal and regulatory constraints. As an example, certain clients may request to receive notifications when research on specific securities is published, and certain clients may request that specific data underlying analysts' fundamental analysis available on our internal client websites be delivered to them electronically through data feeds or otherwise. No change to an analyst's fundamental research views (e.g., ratings, price targets, or material changes to earnings estimates for equity securities), will be communicated to any client prior to inclusion of such information in a research report broadly disseminated through electronic publication to our internal client websites or through other means, as necessary, to all clients who are entitled to receive such reports.

All research reports are disseminated and available to all clients simultaneously through electronic publication to our internal client websites. Not all research content is redistributed to our clients or available to third-party aggregators, nor is Goldman Sachs responsible for the redistribution of our research by third party aggregators. For research, models or other data related to one or more securities, markets or asset classes (including related services) that may be available to you, please contact your GS representative or go to <https://research.gs.com>.

Disclosure information is also available at <https://www.gs.com/research/hedge.html> or from Research Compliance, 200 West Street, New York, NY 10282.

© 2026 Goldman Sachs.

You are permitted to store, display, analyze, modify, reformat, and print the information made available to you via this service only for your own use. You may not resell or reverse engineer this information to calculate or develop any index for disclosure and/or marketing or create any other derivative works or commercial product(s), data or offering(s) without the express written consent of Goldman Sachs. You are not permitted to publish, transmit, or otherwise reproduce this information, in whole or in part, in any format to any third party without the express written consent of Goldman Sachs. This foregoing restriction includes, without limitation, using, extracting, downloading or retrieving this information, in whole or in part, to train or finetune a machine learning or artificial intelligence system, or to provide or reproduce this information, in whole or in part, as a prompt or input to any such system.